Public acceptability of conservation entrance fees: How does reallocating revenues shift support before and after policy implementation?

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Abstract

Entrance fees are widely used to finance conservation and manage tourism in protected areas, yet their implementation often faces strong opposition, threatening ecosystems and local livelihoods sustainability. Evidence on mechanisms that increase public acceptability and on how support evolves following policy implementation remains understudied. We adopt a mixed-methods approach to study the public acceptability of conservation entrance fees, before, shortly after, and nine months after the first fee increase in the Galapagos in 25 years. Qualitative analysis identifies narratives shaping entrance fees support, including views on tourism growth, institutional trust, distributive fairness, and community needs. Estimations based on three survey waves show that, despite initial and persistent opposition among stakeholders, reallocating revenues toward publicly preferred areas substantially and increasingly improves acceptability post-policy implementation. We discuss how transparent and participatory processes can enhance the political feasibility of contested policies, building long-term support with lessons for the governance of other biodiversity-rich destinations.

Keywords: public support, tourism taxes, event study, individual preferences, tourism governance, biodiversity conservation

JEL Classification: H23, Q26, Q57.

1 Introduction

Protected areas are key strategies for addressing the biodiversity loss crisis, while also offering economic opportunities through nature-based tourism (Convention on Biological Diversity, 2022). However, tourism can intensify environmental pressures through negative externalities such as habitat disturbance and wildlife disruption (Monz et al., 2013). Entrance fees are policy tools designed to regulate visitor flows and mitigate these pressures by generating revenues for sanitation, conservation, and protected area management (Chung et al., 2011). Yet the introduction and adjustment of entrance fees are frequently contested, with opposition driven by concerns about economic impacts, limited participation in decision making, and lack of transparency in revenue allocation (Rosselló-Nadal & Sard, 2026; Zou et al., 2024). These tensions can hinder policy implementation, accelerate environmental degradation, and undermine the ecological and development goals that protected areas aim to achieve in the first place (Castrejón & Defeo, 2023; Zou et al., 2024).

This study adopts a sequential mixed-methods approach to examine whether and how revenue allocation can shift public acceptability of entrance fee policies. Exploiting an event study, we analyze acceptability before, shortly after, and long after an increase in conservation entrance fees in the Galapagos National Park, the first adjustment in 25 years. A qualitative phase explores how residents interpret entrance fees within broader narratives about tourism growth, institutional trust, distributive fairness, and conservation revenues use. Building on these insights, we implement three survey waves and estimate mixed-effects regressions to test whether aligning revenue allocation with stakeholder preferences increases public support and how this effect evolves following policy implementation. Our findings are threefold. First, stakeholder opposition is shaped by perceived lack of transparency, concerns about reinforcing inequalities, and perceptions that conservation revenues do not address basic community needs. Second, although visitors are less sensitive to the specific use of revenues, directing funds toward publicly preferred areas substantially increases acceptability among residents. Third, while opposition among residents remains high and persistent in the baseline, the effect of revenue reallocation strengthens over time, progressively improving support for the policy change. We discuss how incorporating stakeholder preferences through participatory and transparent processes can sustain policy support and strengthen protected area governance in

the long-term, with lessons for other biodiversity-rich destinations where local livelihoods depend on the sustainability of nature-based tourism.

This research contributes to the literature on public acceptability of environmental policies and tourist taxes, particularly entrance fees to protected areas. Social acceptability research examines public support for contested policies and highlights determinants such as emotions, perceived fairness, institutional trust, and effectiveness (Kallbekken & Sælen, 2011; Rodriguez-Sanchez et al., 2018; Wüstenhagen et al., 2007). However, most studies remain descriptive, focusing on correlates of support rather than identifying mechanisms capable of shifting initial levels of policy acceptability. In parallel, research on entrance fees has largely focused on demand elasticity and estimations of willingness-to-pay among visitors, with limited attention to the perspectives of residents despite their relevance for participatory governance and effective policy implementation (Daly et al., 2015; King et al., 2025; Mach et al., 2020; Mmopelwa et al., 2007). Taking together, there is a lack of understanding as to what affects the public support for environmental policies,including conservation entrance fees, and how acceptability evolves following real policy implementation (Schuitema et al., 2010; Zou et al., 2024).

Recent work suggests that perceptions of revenue use correlate with support for tourist taxes (Rosselló-Nadal & Sard, 2026), yet evidence on whether allocating revenues to publicly preferred areas can increase acceptability remains scarce. Moreover, existing studies rarely capture dynamic changes in attitudes following policy implementation, particularly over longer time horizons when perceived costs and benefits are realized (Carattini et al., 2018; Mach et al., 2020). Building carefully on these gaps, we address two research questions: (1) Can allocating additional revenues from an entrance fee increase to publicly preferred areas shift initial levels of policy acceptability? and (2) How does this effect change shortly after and long after policy implementation?

This study contributes in three ways. First, it conceptualizes and empirically tests revenue allocation as a mechanism capable of shifting initial levels of public acceptability of entrance fee policies. While prior research documents associations between support and perceptions of revenue use, we directly test whether aligning revenue allocation with stakeholder preferences increases policy support (Rosselló-Nadal & Sard, 2026). Second, the study advances a temporal perspective on policy acceptability by examining how attitudes evolve following actual policy implementation

across multiple time horizons. While Mach et al. (2020) analyze attitudes following the removal of entrance fees over a short period, our study evaluates acceptability of a substantial entrance fee increase across three stages: before, one month after, and nine months after implementation. Third, by integrating narratives on tourism sustainability, distributive fairness, and conservation revenues with a quantitative analysis of stakeholder attitudes, the study provides a comprehensive understanding of how entrance fee policies are evaluated and how policy support can be strengthened through participatory and transparent revenue allocation.

We conduct the study in the Galapagos Islands, the first natural site declared a UNESCO World Heritage Site and a marine biodiversity hotspot of global importance (World Heritage Centre, 2007). With annual visitor numbers approaching 300,000, a seven-fold increase since the 1990s, tourism growth represents a critical concern for ecosystem conservation (Observatorio de Turismo de Galápagos, 2025). Efforts to increase entrance fees were delayed for five years due to local resistance linked to fears about potential economic impacts and dissatisfaction with revenue use (Gobierno del Regimen Especial de Galapagos, 2019). The implementation of the fee increase in August 2024 therefore provides a unique opportunity to examine stakeholder responses across multiple post-implementation horizons. Beyond the Galápagos Islands, the study contributes to broader debates on conservation financing and tourism governance by demonstrating how incorporating stakeholder preferences into revenue allocation can enhance the political feasibility of entrance fee policies.

The remainder of the paper is structured as follows. Section 2 presents the study area and entrance fee policy. Section 3 describes the theoretical framework and empirical specification. Section 4 details the qualitative and quantitative research design. Section 5 presents thematic and econometric findings. Section 6 discusses implications for public acceptability and protected area governance and concludes with directions for future research.

2 Study area and policy change

2.1 Study area

The Galapagos Islands, located in the Eastern Tropical Pacific Ocean approximately 900 km from continental Ecuador, are internationally renowned for their rich biodiversity, hosting more than 2,900 marine species with 18% endemism (World Heritage Centre, 2018). Following Charles Darwin's expedition aboard the *HMS Beagle* in 1835, whose observations inspired the formulation of the theory of evolution, the archipelago has been widely recognized as a natural laboratory of global scientific importance (Grant & Grant, 2008; Nature, 1935). Comprising 19 major islands and more than 100 islets and rocks, Galapagos is currently home to approximately 30,000 residents concentrated on four inhabited islands. Governance operates under the Special Regime Law for the Galápagos, which imposes strict residency rules, regulates terrestrial and marine tourism, and establishes conservation entrance fees for visitors, from which residents are exempt (CGREG, 2015).

Designated a UNESCO World Heritage Site in 1978, the Galapagos were placed on the List of World Heritage in Danger in 2007 due to concerns related to invasive species, overfishing, and unsustainable tourism growth (World Heritage Centre, 2007). Although removed from the list in 2010, concerns about its rapid tourism expansion persist. In 2018, the Ecuadorian government committed to a "zero-growth" tourism model aimed at mitigating the increasing pressures of overtourism (World Heritage Centre, 2018). Yet tourism continues to expand: by 2024, annual visits reached 279,200, with domestic tourists accounting for 45% and foreign visitors for 55%, mainly from the United States and the United Kingdom (Fig. 1). The current visits level is seven times higher than in the 1990s and nearly ten times the resident population (Observatorio de Turismo de Galápagos, 2025; Pizzitutti et al., 2017).

Their global importance, strong tourism dependence, and persistent governance tensions make the Galapagos Islands a relevant setting for examining the political feasibility of key policy instruments, particularly entrance fees designed to balance conservation goals and local development.

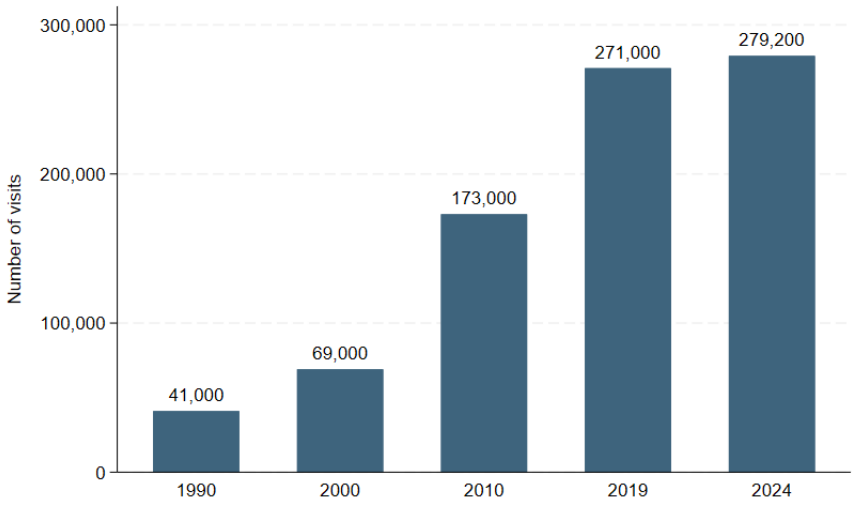

Figure 1: Number of tourists visiting the Galapagos National Park, 1990-2024
Source: Observatorio de Turismo de Galápagos (2025) and Pizzitutti et al. (2017)

2.2 Entrance fee policy

Entrance fees have been the main instrument to manage visitor flows and finance conservation in the Galapagos, but changes to the fee scheme have been highly contentious. Since their introduction in 1998, fees remained unchanged for 25 years with community resistance, demonstrations, legal disputes, and concerns over revenue use repeatedly delaying any changes [1]. After a first attempt five years earlier, in February 2024, the authorities announced the entrance fee adjustment: fees increased from US$6 to US$30 for domestic visitors (a 400% increase) and from US$100 to US$200 for foreign visitors (a 100% increase), effective on August 1, despite persistent opposition among different stakeholders (Consejo de Gobierno del Régimen Especial de Galápagos, 2024; King et al., 2025).

Beyond regulating visits, the new fee scheme aims to generate resources to offset the effects of inflation since its introduction, which has risen the costs of managing and monitoring its terrestrial and marine protected areas. The long-term success of the new entrance fee scheme, however, is more likely with public support. For a fragile ecosystem like the Galapagos, avoiding the sustainable

[1] Social backlash and legal disputes surrounding the increase of Galapagos entrance fees include local congressmen and municipalities challenging the policy change in court, leading to constitutional actions aimed at delaying or avoiding its enforcement (El Diario, 2025). Street demonstrations against the fee increase were often shared on social media, while the research team documented one of these events during fieldwork. International media has also covered the change in Galapagos entrance fee since the first attempt in 2019 to its implementation in 2024 (The Guardian, 2024; The New York Times, 2019).

tourism paradox, where tourism undermine the natural resources they are meant to protect, requires addressing conservation goals together with community livelihoods (Burbano et al., 2022).

3 Theoretical framework and empirical specification

We develop a theoretical framework, informed by the literature on public acceptability, environmental taxes, and visitor fees, to describe how individuals form their attitudes toward an entrance fee policy over time (Chung et al., 2011; Kallbekken & Sælen, 2011; Schuitema et al., 2010; Zhang et al., 2023).

Following Kallbekken and Sælen (2011) on environmental taxes, individuals evaluate a policy change based on their beliefs about the policy consequences. In our setting, this includes two main elements: the expected costs and the expected benefits derived from the policy change. For an entrance fee increase, expected benefits include improvements in ecological quality and public services resulting from additional government investment. Expected costs differ across stakeholder groups. For tourists, the cost is direct: the fee paid to access the protected area. For residents (who are exempt from paying the fee), costs are indirect and depend on the expected economic impact of changes in tourist demand. The expected utility of the policy under scenario s at time t can be expressed as (Eq. 1):

$$\tilde{U}_{t,s} = \underbrace{u_p(p_{t,s}) + u_e(e_{t,s})}_{\text{Public Benefits (+)}} + \underbrace{u_c(c_{t,s})}_{\text{Private cost (-)}} \tag{1}$$

where $p_{t,s}$ and $e_{t,s}$ represent public services and ecological quality, respectively, and $c_{t,s}$ denotes private costs, such as the direct fee paid by tourists and the impact on tourist demand for residents. The value functions $u_p(\cdot)$ and $u_e(\cdot)$ are increasing and capture the policy benefits, while $u_c(\cdot)$ is decreasing and concave in costs. An individual's latent acceptability of the policy change is influenced by the expected utility derived from such a policy change.

The mechanism tested in this study is that the allocation of public revenues following individuals' preferences affects their expectations of the benefits derived from the policy, consequently shifting the level of acceptability. Specifically, we hypothesize that directing additional revenues toward publicly preferred areas increases expected benefits (e.g., improved public services and eco-

logical quality aligned with public needs) while holding private costs constant, i.e., the fee amount paid by visitors and the residents' expected impact on tourism demand remain unchanged. As expected benefits increase in the revenue reallocation scenario ($s = 1$), the acceptability of the policy change is hypothesized to be higher than in the baseline scenario without reallocation ($s = 0$) (Eq. 2):

$$\uparrow u_p(p_{t,s=1}) + u_e(e_{t,s=1}) \Rightarrow Acceptability_{t,s=1} > Acceptability_{t,s=0} \tag{2}$$

Over time, as the policy is implemented and its outcomes become observable, perceptions of costs and benefits update, thereby altering the acceptability of the policy. We hypothesize the following dynamics:

- *Before policy × No reallocation (baseline)*: In the pre-implementation period where no changes in revenue allocation are proposed, acceptability is hypothesized to be low if (ex-ante) expected private costs outweigh expected public benefits, i.e., the disutility from potential costs dominates.

- *After policy (short-term)*: In the immediate aftermath ($t =$ short-after), policy outcomes are not fully realized. If initial expected costs were dominant, ex-post costs expectations are presumed to remain dominant in the short term, leading to persistently low acceptability.

- *If reallocated (revenue reallocation scenario)*: The allocation of revenues according to public's preferences ($s = 1$) shifts the balance of the utility that is expected to be derived from the policy by raising the value of public benefits relative to private costs, thereby increasing acceptability compared to the baseline ($s = 0$).

- *After policy (short-term) × If reallocated*: Combining revenue reallocation ($s = 1$) with the policy implementation exposure ($t =$ short-after) is expected to reinforce the value that individuals assign to the potential benefits derived from the policy if revenues are allocated according to their preferences; this leads to an amplified increase in acceptability.

- *After policy (long-term) × If reallocated*: Over a longer period ($t =$ long-after), realized outcomes reinforce or attenuate initial expectations about policy outcomes. If observed economic impacts (ex-post perceptions) are weaker than expected, acceptability may rise

even in the baseline scenario. If perceived costs persist, reallocating revenues ($s = 1$) may further increase support as it becomes a compensation for realized losses.

Residents' expectations about policy costs depend on their beliefs about the price elasticity of tourist demand: $\epsilon = -\frac{\partial Q}{\partial P} \cdot \frac{P}{Q}$, where Q represents the quantity demanded and P is the price of the environmental good, i.e., the fee to access a protected area (Rosselló-Nadal & Sard, 2026). If residents believe demand to be elastic ($\epsilon > 1$), they anticipate a disproportionate reduction in tourist arrivals following the entrance fee increase, and therefore expect a negative impact on the local economy. Following this, perceptions of costs and benefits can vary with individual characteristics such as economic sector, income, environmental attitudes, and trust in public institutions. We use these variables as control in the econometric specifications.

3.1 Empirical specification

We estimate a mixed-effects ordered logit model in which latent acceptability $Acceptability^*_{s,i}$ depends on exposure to policy implementation, revenue allocation, and their interaction, as shown in Eq. (3):

$$\begin{aligned} Acceptability^*_{is} = \delta_1 AfterShort_i + \delta_2 Reallocation_s \\ + \delta_3 AfterShort_i \times Reallocation_s + \beta X_i + \mu_i + \varepsilon_{is} \end{aligned} \tag{3}$$

where $AfterShort_{is}$ equals 1 if the acceptability of individual i was elicited shortly after policy implementation and 0 otherwise (before policy); $Reallocation_s$ equals 1 if the policy is presented in terms of allocating revenues according to public's preferences, and 0 if presented under the baseline scenario (without revenue reallocation). The interaction term captures how the effect of reallocating revenues evolves in the short-term after policy implementation. X_i includes covariates such as demographics, trust in public institutions, and environmental attitudes. μ_i is a random intercept, accounting for acceptability being measured for the same individual under both revenue use scenarios, and $\epsilon_{s,i}$ is the error term.

To evaluate long-term dynamics, we extend the model by including the variable $AfterLong_i$, which equals 1 if the acceptability of individual i was elicited long after policy implementation and 0 otherwise; we also include its interaction with the variable $Reallocation_{s,i}$. The model is then

defined by Eq. (4):

$$\begin{aligned} Acceptability^*_{is} = \delta_1 AfterShort_i + \delta_2 AfterLong_i + \delta_3 Reallocation_s \\ +\delta_4 AfterShort_i \times Reallocation_s + \delta_5 AfterLong_i \times Reallocation_s \\ +\beta X_i + \mu_i + \varepsilon_{is} \end{aligned} \quad (4)$$

The observed dependent variable is measured on a five-point Likert scale ranging from Strongly Oppose (1) to Strongly Favor (5) as in Rodriguez-Sanchez et al. (2018), serving as an ordered proxy of the latent acceptability variable (Williams & Quiroz, 2020). Given the ordinal outcome and repeated-measures design, we estimate mixed-effects ordered logit models (Davidian & Giltinan, 2003). Residents and tourists are analyzed separately to account for differences in the dynamics of their attitudes over time, as they are exposed to policy costs and policy implementation differently.

4 Research design and methods

4.1 Mixed-methods design

We exploit an event study generated by the increase in conservation entrance fees at the Galápagos National Park, implemented on August 1, 2024. This event provides an opportunity to examine how public acceptability evolves as stakeholders are exposed to a actual policy change.

We employ a sequential mixed-methods design. The qualitative phase is used to understand how residents conceptualize the entrance fee policy, how they assess its potential costs and benefits, and which views and conditions (e.g., transparency, institutional capacity) shape public acceptability. The quantitative phase then tests whether directing additional revenues toward publicly preferred uses increases acceptability, and whether this effect changes before, shortly after, and long after policy implementation. In results, we first present qualitative themes that capture residents' narratives around tourism, conservation, and entrance fee governance, and then test the corresponding mechanisms econometrically using a within-subject survey design and three survey waves among residents and visitors.

4.2 Qualitative study

We conducted six focus groups and six semi-structured interviews with residents ($n = 26$) in Santa Cruz, Galápagos, in June 2024 . Participants were purposively selected to represent sectors with distinct exposure to tourism and conservation governance, including tourism and hospitality, artisanal fishing, agriculture, NGOs, research, and local government. With participants' consent, sessions were audio-recorded and transcribed verbatim; when audio-recording was not possible, detailed notes were taken during and immediately after the session. Discussions were conducted in Spanish, and transcripts were translated into English by bilingual team members.

We analyzed the qualitative data using a hybrid inductive–deductive thematic analysis. Transcripts were first reviewed for familiarization and open-coded to capture statements related to tourism sustainability, transparency, equity, and public concerns about revenue use. Across iterations, we compared and refined codes and clustered them into analytical categories guided by the literature and our theoretical framework. We organized categories into four higher-order themes: (i) Tourist Demand and Sustainable Growth; (ii) Trust and Transparency; (iii) Fairness and Inequality; and (iv) Conservation Revenues and Community Needs. We developed a codebook including 29 analytical codes nested within these themes and applied it to the dataset. Quotes are identified using anonymized participant IDs, with role descriptors included where relevant.

Insights from this phase informed the design of the survey instrument, including development of the baseline and revenue reallocation scenarios; list of revenue allocation areas used in the preference elicitation task; and identification and validation of key constructs included in the survey and econometric analysis (e.g., institutional trust, environmental attitudes, economic sector). The qualitative findings also guided the interpretation of quantitative results.

4.3 Quantitative study

We conducted on-site surveys in three waves using a within-subject design, which elicits multiple acceptability responses from the same individual across policy scenarios (Davidian & Giltinan, 2003; Fuster & Zafar, 2023). The core section of the survey followed three steps. First, respondents evaluated the acceptability of the entrance fee increase under a baseline scenario i.e., without changes

in the allocation of additional revenues. Second, respondents ranked nine potential investment areas for allocating additional funds. These categories reflect legally permissible uses of conservation entrance fees and were validated with protected area managers. The order of items was randomized for each respondent. Third, respondents reassessed the acceptability of the fee increase under a revenue reallocation scenario in which additional revenues were allocated according to their stated preferences. This design identifies how acceptability changes when the policy is explicitly linked to public's preferred uses of revenues.

To study dynamics over time, we conducted three survey waves to compare results in the short term (before vs one month after implementation) and in the long term (a follow-up nine months after implementation). We selected a within-subject design due to its statistical efficiency relative to between-subject designs (Fuster & Zafar, 2023), which was important given the logistical difficulties of fieldwork in the Galápagos and the event-study setting. The design is also policy-relevant, as most stakeholders were aware of the imminent (pre-policy) or recent (post-policy) fee increase.

4.3.1 Survey details

Based on the qualitative findings and the literature, we drafted the questionnaire and pretested it with seven individuals, refining wording and improving the interpretability of the public acceptability section. We then conducted a pilot survey with 26 residents and 22 tourists, leading to further adjustments to question wording and completion time.

At the start of the survey, respondents were informed about the research team, confidentiality protocols, and ethical approval. They were also informed that aggregated results would be shared with policymakers, a practice recommended to strengthen perceived consequentiality in stated preference research to encourage truthful responses (Carson & Groves, 2007). After providing consent, respondents answered warm-up questions on recreational preferences, conservation perceptions, and prior knowledge of the planned or recent entrance fee change. The main section elicited acceptability of increasing entrance fees using a five-point Likert scale ranging from "Strongly Oppose" to "Strongly Favor", as in Rodriguez-Sanchez et al. (2018). Residents were additionally asked about expected impacts of the fee increase on tourist visits and their trust in public institutions (measured on a five-star scale). Tourists reported travel characteristics, including number of pre-

vious visits, length of stay, travel experience, and visit modality (land-based or liveaboard). Both groups completed demographic questions, the New Ecological Paradigm (NEP) battery to measure environmental attitudes (Dunlap et al., 2000; Zhu & Lu, 2017), and an open-ended section for additional comments.

4.3.2 Data collection

We administered 1,358 on-site surveys among residents and visitors. The short-term analysis includes 762 surveys: 325 collected before the policy change (July 2024) and 437 shortly after implementation (August-September 2024). The follow-up includes 596 surveys collected nine months after implementation (May-June 2025). Data were collected by two trained local assistants under the supervision of the research team.

Resident surveys ($n = 629$) were conducted face-to-face in Santa Cruz Island using tablets and the SurveyLab platform. We used quota sampling to approximate the population distribution by gender and age. Given that roughly 60% of local employment is directly or indirectly linked to tourism (e.g., commerce, hospitality, transport), we also aimed to reflect this distribution in the sample (Pizzitutti et al., 2017). Tourist surveys ($n = 729$) were administered in paper format in the departure area of Baltra Airport, the main gateway to the archipelago, using a systematic sampling approach. Field assistants approached travelers seated at every fifth seat; if the selected traveler refused, or was identified as a resident rather than a tourist, the next traveler was selected. Questionnaires were available in Spanish and English, and field assistants provided clarification when needed. Only participants aged 18 or older who provided informed consent were surveyed.

5 Results

5.1 Qualitative findings

The qualitative analysis identified four higher-order themes that help to understand how residents evaluate entrance fee increases and under what conditions they become acceptable: (i) Tourist Demand and Sustainable Growth; (ii) Trust and Transparency; (iii) Fairness and Inequality; and

(iv) Conservation Revenues and Community Needs. Below we use illustrative quotes to describe each theme and highlight how each maps onto the survey design and empirical specification.

5.1.1 Tourist Demand and Sustainable Growth

Debates about tourism growth in the Galápagos Islands reflect tensions between economic dependency and ecological sustainability. Residents interpret the entrance fee policies within broader narratives about the appropriate tourism model for the archipelago. For some participants, particularly those working in tourism, growth was portrayed as economically necessary and structurally difficult to reverse. Tourism was framed as the financial engine of conservation, reinforcing a mutually dependent relationship between environmental protection and increased visitor flows. A tour operator stated:

> "It was obvious that this [tourism growth] would happen. Tourism is needed to finance conservation and a good story always sell better. This brings in more people, requires more labor, and generates income to pay them. This will just continue" (Tour operator, Q27).

From this perspective, entrance fees were viewed less as instruments to regulate tourism growth and more as mechanisms to finance its consequences. In contrast, participants from environmental and fishing sectors highlighted their concerns about exceeding the ecological carrying capacity of the islands and the environmental risks associated with greater visitor numbers. They interpreted fee increases through the lens of long-term sustainability, questioning whether the current tourism model can absorb continued growth without degrading fragile habitats. A tour operator also acknowledged that the economic benefits derived from tourism may not offset long-term environmental risks:

> "The increase in tourists is something positive. But there needs to be greater efforts on protecting the ecosystem we have, because if it is destroyed, later the fall will be drastic" (Tour operator Q33).

Residents also differentiated between visit modalities. Liveaboard tourism was often perceived as economically extractive and weakly embedded in the local economy. While land-based tourism

was associated with both a necessary revenue stream for local businesses and a concerning pressure due to mass tourism and less conscious visitors. In terms of the demand impact of higher entrance fees, most participants believed foreign demand to be relatively price-inelastic, whereas domestic demand was considered more sensitive, although the fee might be still affordable under the new scheme.

These narratives reveal that support for entrance fees is embedded within broader views on tourism growth and mediated by beliefs about demand elasticity. In the quantitative design, these mechanisms are captured through controls for economic sector and environmental attitudes, as well as through questions eliciting residents' expectations and perceptions of the entrance fee impacts on tourist demand. Although the fee scheme is defined by nationality, the tourist survey includes controls for both nationality and visit modality (land-based vs. liveaboard), allowing us to compare attitudes across different tourism segments.

5.1.2 Trust and Transparency

Across interviews and focus groups, institutional trust emerged as a condition for policy acceptability. Even among participants who recognized the need for additional conservation funding, support for the entrance fee increase was contingent upon trust in institutional capacity, transparency, and accountability. Policymaking was several times described as reactive and inconsistent, characterized by "patchwork solutions" and decisions made "hastily." Some residents questioned whether the fee increase was primarily motivated by fiscal needs or international reputation rather than sustainability objectives or local priorities:

> "The decision to increase the fee was made solely as a strategy to collect more funds, and not necessarily to limit tourism, since no [visitor] caps have been established and the new fee will not restrict the growth in visitor numbers." (Natural scientist, Q55).

Concerns about conservation being instrumentalized for tourism promotion further weakened trust in institutions, with skepticism about the motivations behind the policy change and whether revenues would be managed effectively. Transparency in revenue allocation repeatedly emerged as a factor in shaping support:

> ”I would lean more in favor of the increase if there were clarity and certainty about where the funds will go” (Hospitality entrepreneur, Q34).

These narratives suggest that residents evaluate entrance fee policies based on both the expected benefits and on procedural legitimacy. Trust shapes expectations about whether potential benefits will materialize and whether revenues will be managed transparently. This mechanism directly informed the quantitative design. The questionnare measures trust in public institutions and elicits perceived survey consequentiality, allowing us to test whether institutional trust explains the acceptability of increasing entrance fees.

5.1.3 Fairness and Inequality

Perceptions of distributive injustice and unfairness within the tourism economy are also narratives through which residents assess the entrance fee policy. Tourism growth was frequently described as producing asymmetric benefits, with local businesses generating limited profits relative to large tourism operators. Expressions such as “We are only left with the crumbs” (NGO member, Q14) captured concerns about economic inequality. Liveaboard tourism was described as highly profitable yet contributing less to the local economy than land-based tourism. Hospitality entrepreneurs linked this dynamic to intensified competition among small businesses and declining margins:

> “Backpackers generate more income to local businesses [than liveaboard passengers]. We are forced to compete fiercely among ourselves. It’s like a war. This competition is getting out of hand” (Restaurant owner, Q31).

Concerns about fiscal centralization were also frequent. Participants noted that collected revenues are first transferred to the national treasury before being redistributed to local institutions, raising fears that such funds may not fully return to the islands if not executed in a efficient and timely manner. These concerns are consistent with official figures showing that, between 2015 and 2023, the collection of entrance fee revenues increased while public spending by the national park authority declined (Consejo de Gobierno del Régimen Especial de Galápagos, 2024).

Therefore, entrance fees were evaluated through distributive justice considerations: who pays, who benefits, and whether burdens are proportionate. When the policy was perceived as reinforcing

existing inequalities, its legitimacy weakened. These insights informed the quantitative strategy in two ways. First, the econometric specifications control for income and employment sector to capture differential exposure to tourism benefits. Second, the revenue reallocation scenario directly tests whether redistribution toward publicly preferred areas can increase policy acceptability.

5.1.4 Conservation Revenues and Community Needs

Finally, residents assessed the entrance fee policy through its capacity to deliver tangible improvements in everyday life. While conservation funding needs were acknowledged, many participants emphasized deficits in basic services, including as more immediate priorities: safe water, sanitation, waste management, and infrastructure. A dive guide expressed:

> "Health comes first, we need better public services like safe water and sewage systems, where we are currently stagnant. Then infrastructure, not only for visitor sites but also for docks and ports" (Scuba diving leader, Q25).

For many, conservation and social investment were complementary rather than competing goals. However, social needs were often framed as a precondition for strengthening environmental awareness and reducing tensions between residents and conservation authorities. A natural scientist noted:

> "The new fees [scheme] will allow for the collection of more funds that are necessary for different investments, but these should first address the needs of the population. If people's needs are not met, they will not be able to think about conservation" (Natural scientist, Q50).

Overall, these narratives suggest that residents evaluate entrance fees largely through their capacity to deliver tangible benefits; policy legitimacy depended on whether revenues would translate into visible improvements and internalize tourism environmental externalities. This theme directly informed the quantitative study design. The list of potential revenue allocation areas included in the survey was derived from these discussions and subsequently validated with authorities to ensure compliance with legally permissible uses.

5.2 Quantitative results

5.2.1 Sample characteristics

Table 1 reports descriptive statistics for the resident sample ($n = 629$) across survey waves. Gender distribution is balanced with 46-53% males and 47-54% females. Respondents range from 18 to 70 years old, with mean ages between 35 and 37. Average monthly income is approximately USD 970, comparable to the 2021 Galápagos Census figure (USD 978) (INEC, 2023, 2024). In terms of education, 44–63% completed secondary education or lower, and 37–56% hold undergraduate or postgraduate degrees. Employment profiles reflect the tourism-based economy: 44–50% work directly in tourism (e.g., tour agencies, hospitality, transport), 15–19% in commerce, 10–12% in government or conservation (including NGOs and research), and 7–11% are students or unemployed. Most residents reported being aware of the policy change. Statistical tests indicate no differences in age across waves, while income differs across at least one wave at the 5% level; gender and economic sector are not statistically different between waves. For comparability in the econometric analysis, we use entropy balancing weights to match key characteristics across survey waves, including gender, age, education level, economic sector, and income. Balanced sample characteristics are reported in the Appendix (Table A.1).

Table 2 summarizes tourist sample characteristics ($n = 729$). Across waves, the sample is approximately split between foreign and domestic (Ecuadorian) visitors. Tourists range from 18 to 80 years old, with mean ages between 39 and 43. In terms of visit modality, 67–79% stayed land-based and 21–33% traveled on liveaboard cruises. No statistical differences are detected across waves for gender, income, occupation, or length of stay. The final wave includes relatively more foreign and highly educated tourists and a lower share of land-based visitors; tourists interviewed after implementation were also more aware of the fee increase. For econometric analysis, we apply entropy balancing to match demographics and visit modality (land-based or liveaboard) across waves. Balanced tourist characteristics are reported in the Appendix (Table A.2).

Table 1: Sample characteristics - Local Residents

Variable		Before policy change	After policy one-month	After policy nine-month	Ho: No difference p-value
Residence Status	Non-permanent	14%	12%	10%	0.355
	Permanent	86%	88%	90%	
Gender	Male	51%	46%	53%	0.287
	Female	49%	54%	47%	
Age 18 - 70 years	Mean	37.4	36.3	34.8	0.062
	(std. dev.)	(13.5)	(10.6)	(10.8)	
Education level	Secondary or lower	55%	44%	63%	<0.001
	Higher education	45%	56%	37%	
Monthly income USD 0.25k - 6k	Mean	1.09k	0.91k	0.91k	0.019
	(std. dev.)	(0.97)	(0.65)	(0.50)	
Economic sector	Commerce	15%	16%	19%	0.501
	Tourism-related	46%	50%	44%	0.406
	Govt. or conservation	10%	12%	12%	0.871
	Other sectors	23%	17%	14%	0.074
	Students and none	6%	5%	11%	0.026
Policy change awareness	Yes	97%	99%	89%	–
	No	3%	1%	11%	
Observations	–	203	210	216	N = 629

Notes: Survey waves are compared using analysis of variance (ANOVA). The null hypothesis (Ho) states that all group means are statistically equal.

Table 2: Sample characteristics - Tourists

Variable		Before policy change	After policy one-month	After policy nine-months	Ho: No difference p-value
Tourist nationality	Domestic	61%	57%	42%	<0.001
	Foreign	39%	43%	58%	
Gender	Male	41%	47%	50%	0.179
	Female	59%	53%	50%	
Age 18 - 83 years	Mean	42.7	38.8	41.1	0.034
	(std. dev.)	(15.1)	(14.4)	(13.7)	
Education	Secondary or lower	23.3%	16.3%	11.8%	0.008
	Higher education	76.7%	83.7%	88.2%	
Monthly income USD 0.3k - 150k	Mean	12.5k	8.3k	10.5k	0.388
	(std. dev.)	(33.0k)	(23.2k)	(27.2k)	
Occupation	Private worker	56%	56%	62%	0.323
	Public employee	25%	22%	24%	0.766
	Retired	11%	7%	7%	0.382
	Other	8%	15%	7%	0.010
Tourist modality	Land-based	79%	73%	67%	0.022
	Liveaboard	21%	27%	33%	
Lenght of stay 3 - 60 days	Mean	7.8	6.9	7.3	0.461
	(std. dev.)	(9.7)	(5.2)	(6.2)	
Policy change awareness	Yes	48%	62%	38%	<0.001
	No	52%	38%	62%	
Observations	–	122	227	380	N = 729

Notes: Survey waves are compared using analysis of variance (ANOVA). The null hypothesis (Ho) states that all group means are statistically equal.

5.2.2 Descriptive results

Table 3 reports residents' acceptability of increasing entrance fees under the baseline scenario (no revenue reallocation). Support is low: 7% and 10% of residents strongly support increasing entry fees for national and foreign tourists, respectively, while 17% and 14% remain indifferent. By contrast, around 51% strongly oppose increases for national tourists and 49% strongly oppose increases for foreign tourists. Overall, residents do not differentiate between national and foreign fee increases.

Table 3: Public acceptability - Pearson's chi-squared test: National fee increase and Foreign fee increase - Respondents: Residents

Public acceptability 5-point Likert scale	National fee increase	Foreign fee increase	Difference %	Pearson's chi-squared Ho: No difference
Strongly oppose	51%	49%	-2%	chi2 = 6.977
Somewhat oppose	16%	15%	-1%	p = 0.137
Neutral	17%	14%	-3%	
Somewhat favor	9%	12%	3%	
Strongly favor	7%	10%	3%	
Observations	629	629	-	-

Table 4 summarizes tourists responses. In contrast to residents, the attitudes of tourists are more evenly distributed. Approximately 15% and 13% of respondents strongly oppose fee increases for national and foreign visitors, respectively, while 29% and 22% remain indifferent, and 21% and 27% strongly support the increases. Tourists thus exhibit greater acceptability of higher fees than to residents and, unlike residents, express greater support for higher fees for foreign visitors.

Table 4: Public acceptability - Distribution test - Pearson's chi-squared test: National fee increase and Foreign fee increase - Respondents: Tourists

Public acceptability 5-point Likert scale	National fee increase	Foreign fee increase	Difference %	Pearson's chi-squared Ho: No difference
Strongly oppose	15%	13%	-2%	chi2 = 15.685
Somewhat oppose	17%	17%	0%	p = 0.003
Neutral	28%	21%	7%	
Somewhat favor	19%	22%	3%	
Strongly favor	21%	27%	6%	
Observations	725	728	-	-

Respondents also stated their preferences for the use of additional revenues generated by the

entrance fee increase by ranking investment areas from a list provided in the survey. Rankings were aggregated using the Rank Order Centroid method (Table 5). Among residents, the highest priority is investment in safe water and wastewater systems, followed by species conservation, improvements to tourist sites, infrastructure in populated areas, and solid waste management. Tourists place the highest priority on species conservation, followed by solid waste management, safe water and wastewater systems, protected area monitoring, and invasive species control. These rankings define the reallocation scenario used later in the survey and reflect the importance placed to tangible policy benefits shown in the qualitative analysis, particularly among residents.

Table 5: Preferences for allocating additional revenues from the entrance fee increase - Rank Order Centroid (ROC) Method - Residents and Tourists

Residents			Tourists		
Rank	Areas for revenue use	Score	Rank	Areas for revenue use	Score
1st	Water and wastewater system	84.6	1st	Species conservation	122.1
2nd	Species conservation	78.3	2nd	Solid waste management	90.3
3rd	Improvement of tourist sites	65.2	3rd	Water and wastewater system	80.7
4th	Infrastructure in populated areas	64.2	4th	Protected areas management and monitoring	73.7
5th	Solid waste management	60.6	5th	Control of invasive species	73.3

Figures 2 and 3 illustrate how residents acceptability changes across scenarios and survey waves. For national fees, 48% of residents strongly opposed the increase before implementation under the baseline scenario. Under revenue reallocation, strong opposition falls to 29%, with more than one-third of initial opponents shifting toward support when revenues are allocated to their preferred areas. This shift becomes more pronounced after policy implementation: opposition under the reallocation scenario declines further after one month and nine months, while opposition under the baseline scenario remains high (48% after one month) and increases to 56% after nine months. Similar patterns hold for foreign fees. Overall, residents willingness to shift toward support seems highly influenced by whether revenues are allocated in line with public priorities.

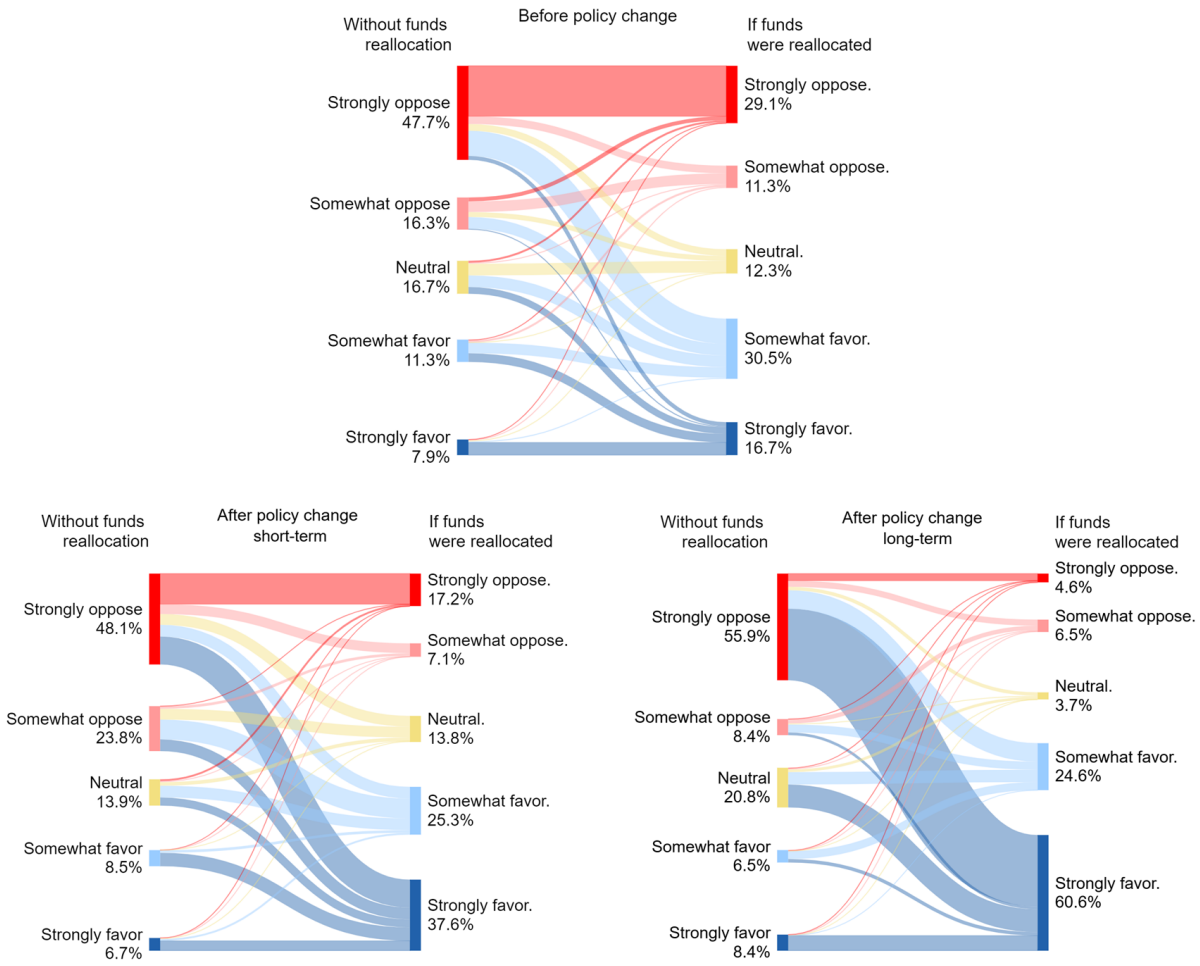

Figure 2: Public acceptability - National fee increase
Before policy change, after short-term and after long-term - With and without revenues reallocation. Respondents: Residents

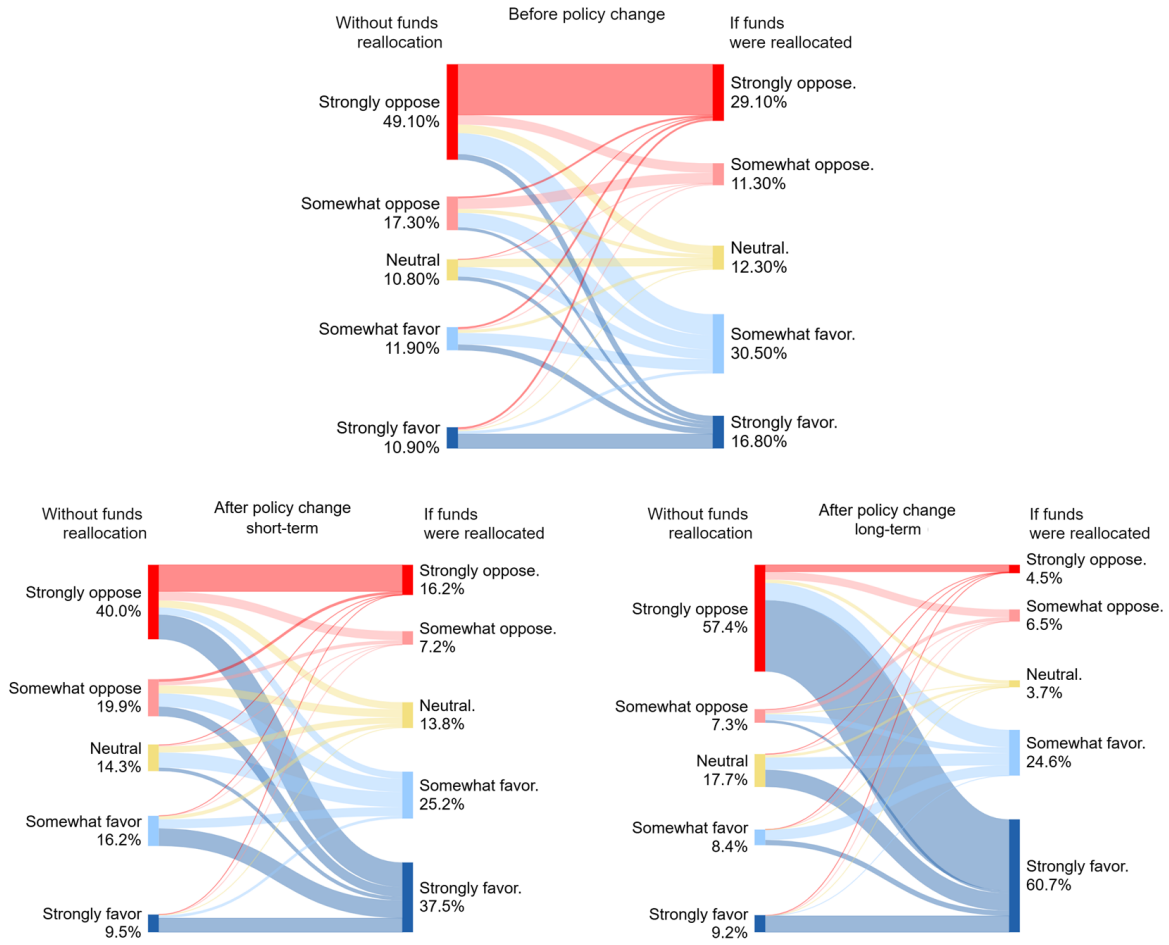

Figure 3: Public acceptability - Foreign fee increase
Before policy change, after short-term and after long-term - With and without revenues reallocation. Respondents: Residents

Figures 4 and 5 show similar patterns for tourists, separated by nationality. National tourists display stronger opposition before implementation, but this softens after enforcement, with responses shifting toward neutrality. Foreign tourists remain consistently neutral or supportive (over 70%) both before and after implementation, under both scenarios. When comparing scenarios, national tourists show a clearer positive shift in support under reallocation, while foreign tourists (already largely supportive) show only modest changes. Tables A.3 to A.7 in the appendix reports further statistical tests for residents and tourists.

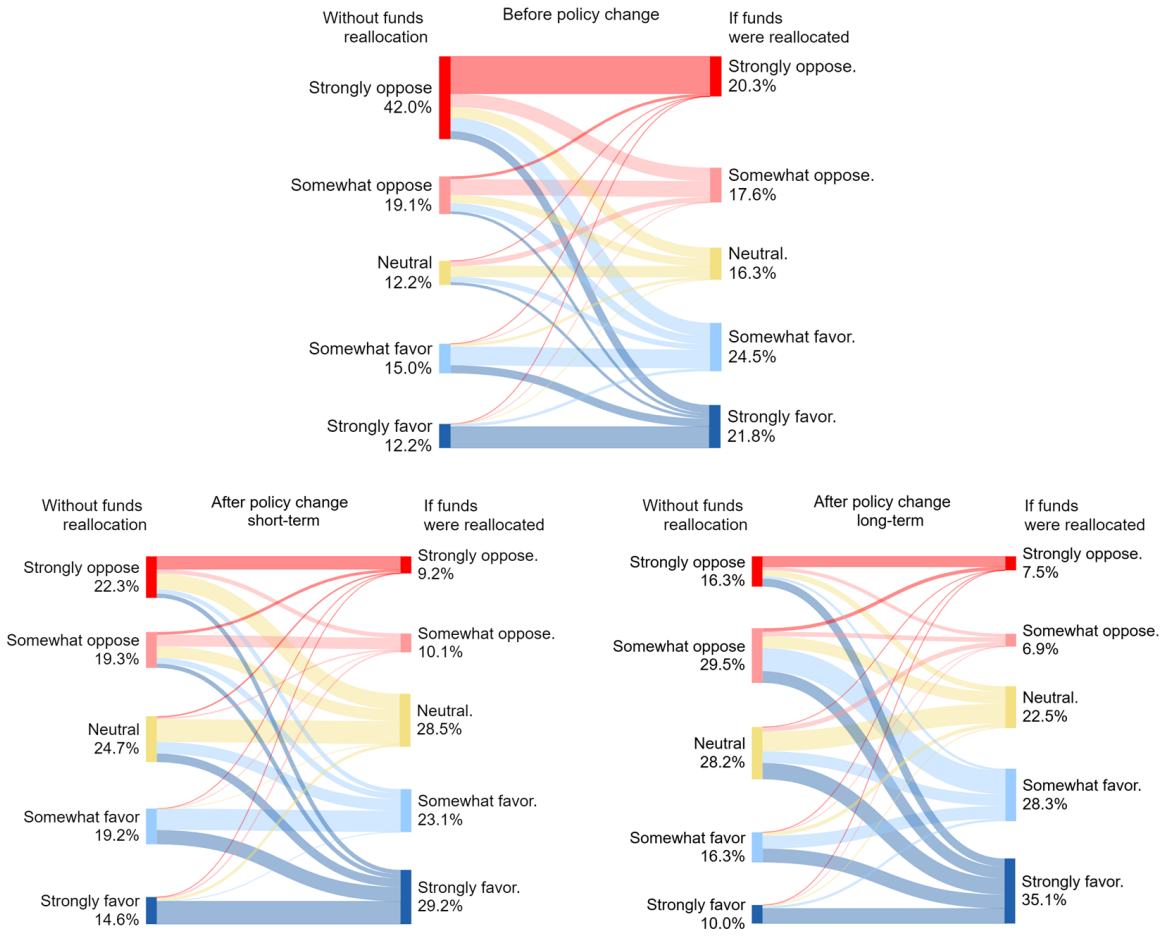

Figure 4: Public acceptability - National fee increase
Before policy change, after short-term and after long-term - With and without revenues reallocation. Respondents: National Tourists

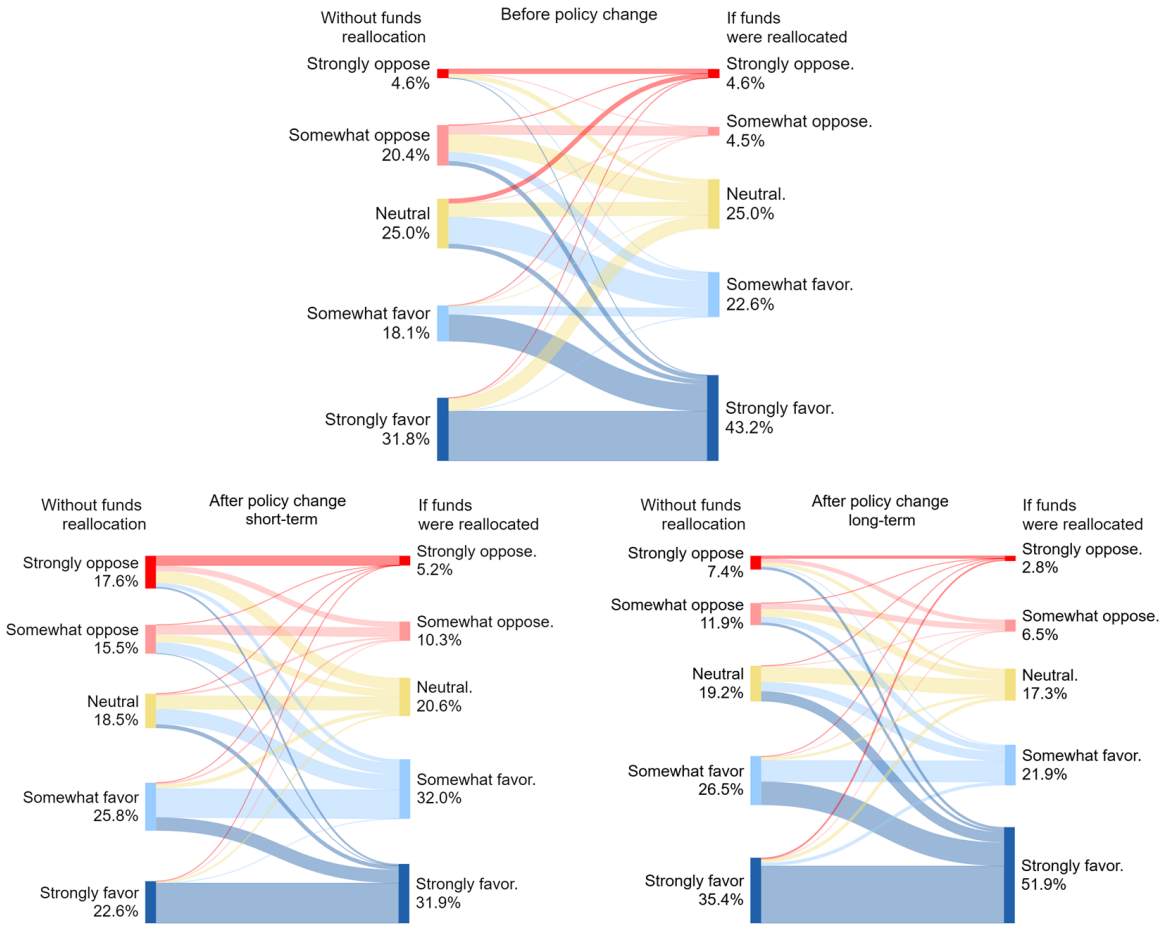

Figure 5: Public acceptability - Foreign fee increase
Before policy change, after short-term and after long-term - With and without revenues reallocation. Respondents: Foreign Tourists

5.2.3 Econometric results: short-term

We estimate mixed-effects ordered logit models of public acceptability (five-point Likert scale). Specifications (1)–(2) assess support of increasing entrance fees for national tourists and specifications (3)–(4) for foreign tourists. Key predictors include the variable indicating whether the policy change has been implemented (*before policy*, *after short-term*), the scenario that modifies the allocation of additional revenues (*no reallocation*, *if reallocated*), and their interaction. The baseline category for comparison is *before policy, no reallocation*. Specifications (2) and (4) additionally include demographic characteristics, trust in public institutions, environmental attitudes, and perceived survey consequentiality.

Figure 6 presents the estimated coefficients and 95% confidence intervals for the residents in the short term. Under the baseline scenario, residents' acceptability is not statistically different shortly after implementation, compared to before policy implementation. By contrast, revenue reallocation significantly increases acceptability, consistent with the mechanism that aligning revenues with public preferences increases expected benefits. The interaction term indicates an additional increase in acceptability shortly after implementation under reallocation. According to specifications (2) and (4), trust in public institutions is positively associated with residents support for increasing entrance fees for both national and foreign visitors. Stronger pro-environmental attitudes, higher income, and employment in government or conservation are also positively associated with public acceptability. Coefficients for age, gender, and education are not statistically different from zero. These results align with qualitative evidence that policy support depends on perceived institutional credibility and tangible public benefits. The estimations are robust to not using entropy balancing to match survey waves characteristics. Full regression tables are reported in the Appendix (Table A.8).

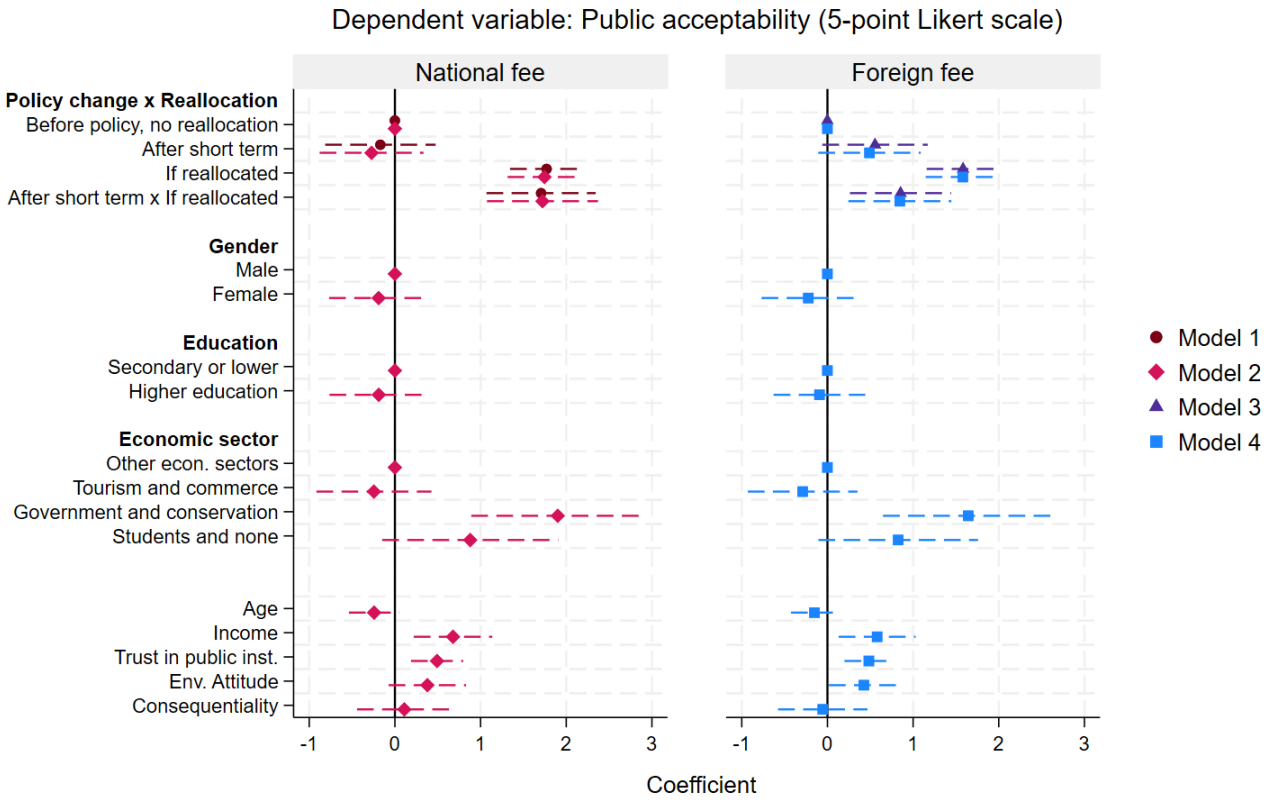

Figure 6: Mixed Effects Ordered Logit - Short-term results - Respondents: Residents

Notes: This figure presents estimated coefficients and 95% confidence intervals (dashed lines) from mixed-effects ordered logit models of residents' support for increasing entrance fees to the Galápagos National Park, before and shortly after (one-month) policy implementation. Models 1–2 refer to support for increasing fees for domestic tourists, while Models 3–4 refer to fees for foreign tourists. Models 2 and 4 controls for demographics, trust in public institutions, environmental attitudes, and perceived survey consequentiality. Regression tables are reported in the Appendix, Table A.8.

Figure 7 shows the regression coefficients and confidence intervals for tourists respondents in the short-term. As for residents, revenue reallocation significantly increases acceptability, though coefficients are smaller and estimated with wider confidence intervals. No significant differences are evidenced between pre-policy and post-policy (short-term) periods under either scenario. With controls (specifications 2 and 4), domestic land-based tourists are consistently less supportive than liveaboard tourists and foreign visitors. Higher education and stronger pro-environmental attitudes are positively associated with public acceptability. Perceived survey consequentiality also increases support among tourists, unlike among residents. Age, gender, and income are uncorrelated to acceptability, likely because the effects of income are absorbed by controls for tourist nationality, visit modality, and education. The estimations are robust to not using entropy balancing weights. Full regression tables are in the Appendix (Table A.9).

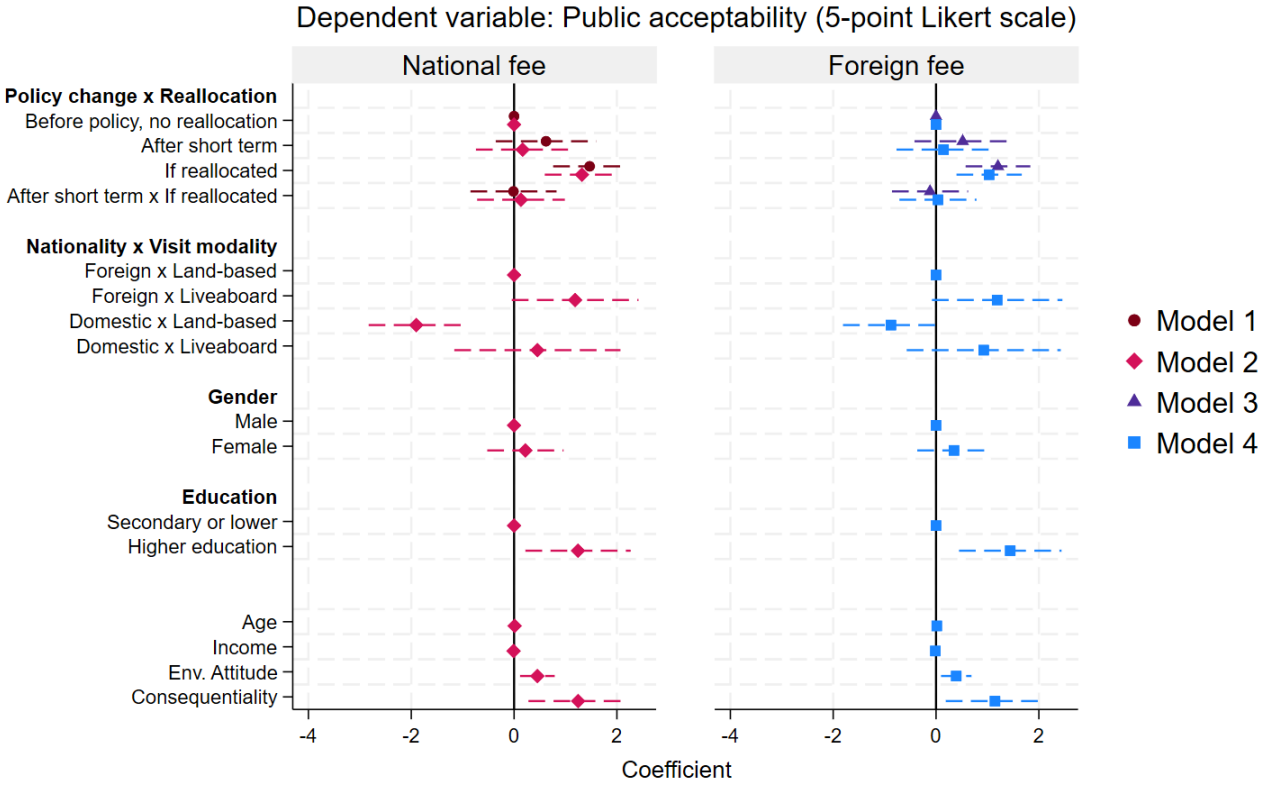

Figure 7: Mixed Effects Ordered Logit - Short-term results - Respondents: Tourists
Notes: This figure presents estimated coefficients and 95% confidence intervals (dashed lines) from mixed-effects ordered logit models of tourists' support for increasing entrance fees to the Galápagos National Park, before and shortly after (one-month) policy implementation. Models 1–2 refer to support for increasing fees for domestic tourists, while Models 3–4 refer to fees for foreign tourists. Models 2 and 4 controls for demographics, visit modality, environmental attitudes, and perceived survey consequentiality. Regression tables are reported in the Appendix, Table A.9.

5.2.4 Econometric results: follow-up

Figure 8 presents estimates for residents nine months after policy implementation, including the short-term period for comparison. Under the baseline scenario, acceptability marginally declines after nine months, significant only in specifications with covariates (Models 2 and 4). This pattern is consistent with respondents perceiving realized costs (e.g., tourism impacts) as comparable to or worse than expected. Survey responses support this interpretation: before implementation, 85% expected a reduction in visits; nine months later, 72% perceived fewer tourists, and two-thirds of this subgroup attributed the reduction directly to the new entrance fee policy (Table 6).

In contrast, the effect of revenue reallocation strengthens over time. Figure 9 illustrates the marginal effects of the interaction terms across the five levels of public acceptability. Compared to the baseline (no reallocation), the reallocation of revenues increases the probability of "Strongly Favoring" the entrance fee policy by approximately 10% before implementation, 30% after one month, and 50% after nine months. At the same time, the reallocation of revenues reduces the likelihood of "Strong Opposition" across these three time horizons. Together, these results suggest that, over a longer term, residents' acceptability becomes increasingly contingent on whether revenues are directed to publicly valued benefits. Full regression tables are reported in the Appendix (Table A.10).

Table 6: Perceptions and expectations of the entry fee increase impact on tourist demand - Respondents: Residents

	Ex-ante expectations Before policy change	Ex-post perceptions After policy long-term
Higher or equal number of tourists	15%	28%
Lower number of tourists	85%	72%
If responded "lower number of tourists"	Before policy change	After policy long-term
Due to the entry fee increase	–	68%
Due to conditions in the mainland	–	29%
Due to other reasons	–	3%

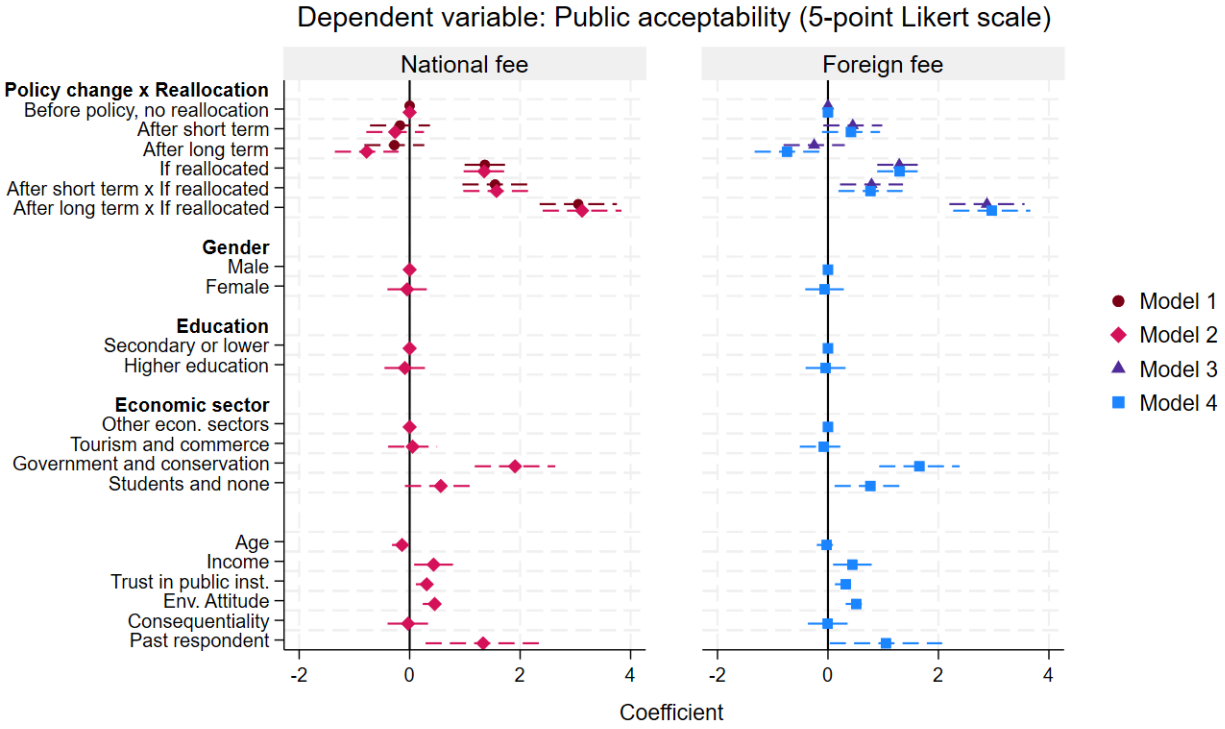

Figure 8: Mixed Effects Ordered Logit - Follow-up study - Respondents: Residents
Notes: This figure presents estimated coefficients and 95% confidence intervals (dashed lines) from mixed-effects ordered logit models of residents' support for increasing entrance fees to the Galápagos National Park, before, one month after and nine months after policy implementation. Models 1–2 refer to support for increasing fees for domestic tourists, while Models 3–4 refer to fees for foreign tourists. Models 2 and 4 controls for demographics, trust in public institutions, environmental attitudes, and perceived survey consequentiality. Regression tables are reported in the Appendix, Table A.10.

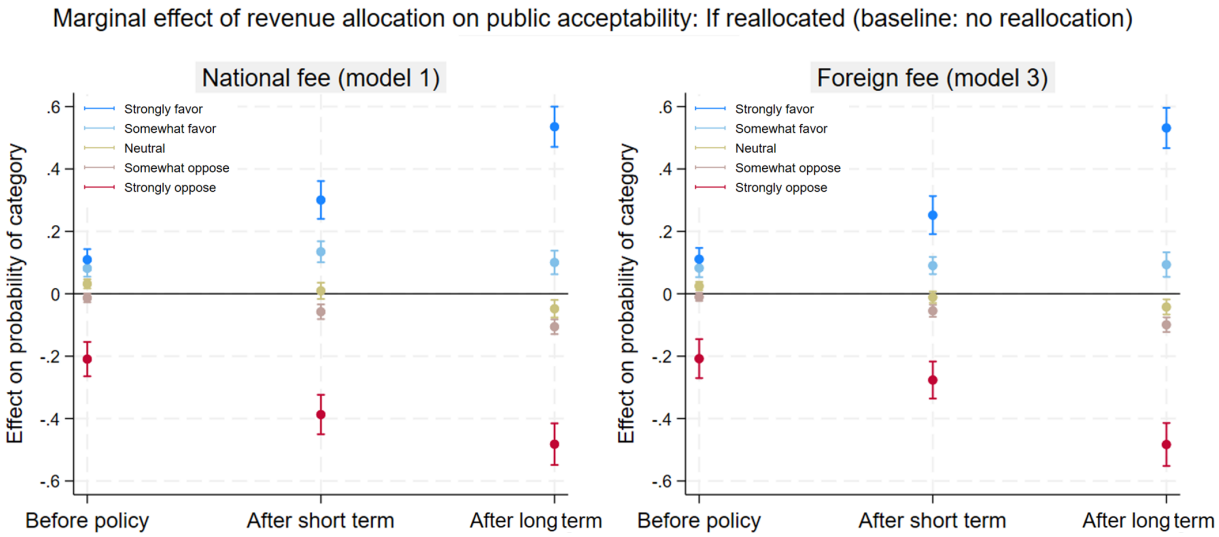

Figure 9: Marginal effects of revenues reallocation on public acceptability - Follow-up study - Respondents: Residents

Notes: This figure reports the marginal effect of the revenue reallocation on residents' acceptability of increasing entrance fees to the Galápagos National Park, compared to the baseline scenario (no reallocation) across three periods: before, shortly after (one-month) and long after (nine-months) policy implementation. Estimates are based on Models 1 and 3, corresponding to increases in fees for domestic and foreign tourists, respectively. Regression tables are reported in the Appendix, Table A.10.

Figure 10 illustrates estimates for tourists. Under the baseline scenario, tourists' acceptability increases after nine months, but this effect is not statistically significant once controls are included (Models 2 and 4). The overall reallocation coefficient remains positive and statistically significant across specifications. The interaction between reallocation and the long-term period is statistically different from zero only for increasing fees for national tourists (Models 1 and 2) but not for foreign visitors (Models 3 and 4). Figure 11 illustrates the marginal effects of revenue reallocation across the different levels of public acceptability and policy implementation periods. After nine months of implementation, the reallocation of revenues increases the probability of "Strongly Favoring" higher entrance fees for national tourists by almost 20%. This increase, however, is primarily driven by a reduction in the probability of "Somewhat Supporting" the policy change, with no significant reduction observed in the probability of being neutral or opposing the policy (Model 1). The marginal effect of revenue reallocation remains relatively stable over time when considering the entrance fees for foreign tourists (Model 3). Covariate patterns remain consistent with the short-term analysis: land-based tourists are less supportive than liveaboard tourists, while higher education, stronger environmental attitudes, and higher perceived consequentiality are associated with greater policy support. Full regression tables are reported in the Appendix (Table A.11).

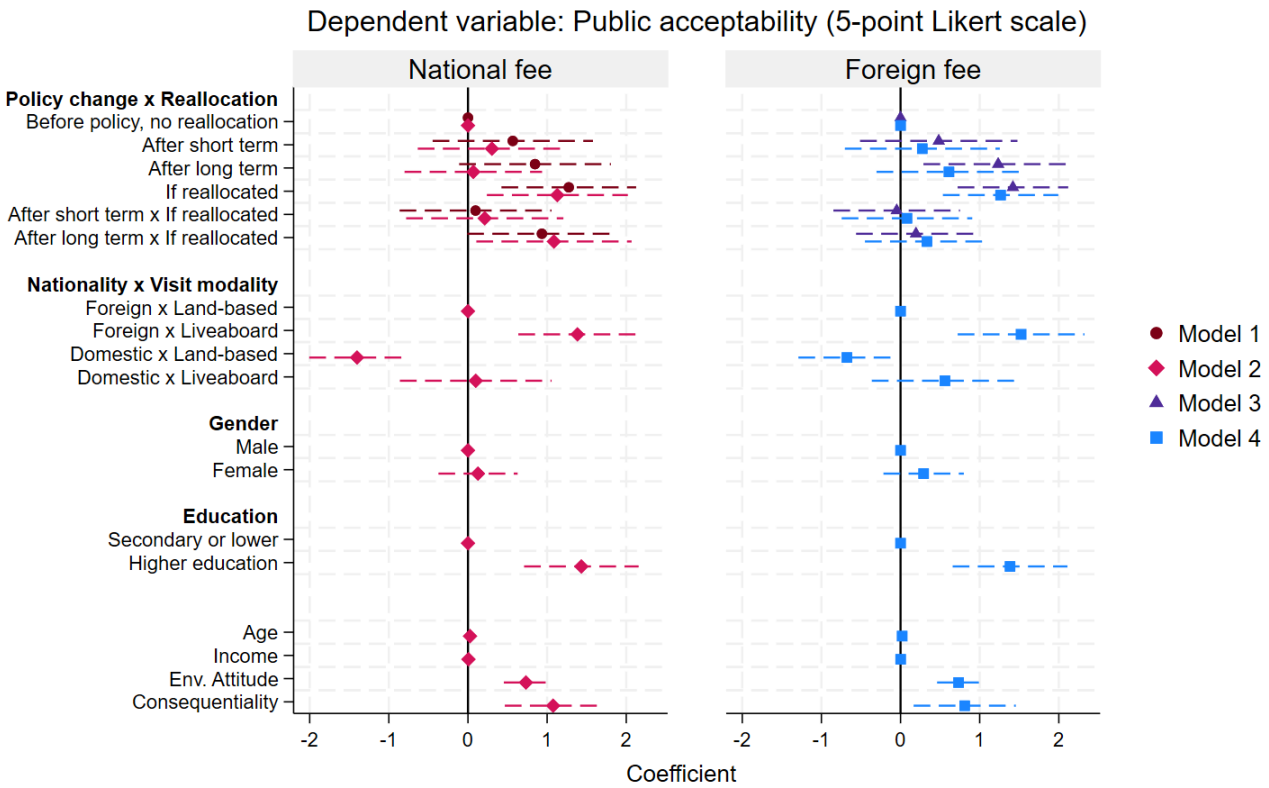

Figure 10: Mixed Effects Ordered Logit - Follow-up study - Respondents: Tourists
Notes: This figure presents estimated coefficients and 95% confidence intervals (dashed lines) from mixed-effects ordered logit models of tourists' support for increasing entrance fees to the Galápagos National Park, before, shortly after (one-month) and long after (nine-months) policy implementation. Models 1–2 refer to support for increasing fees for domestic tourists, while Models 3–4 refer to fees for foreign tourists. Models 2 and 4 controls for demographics, visit modality, environmental attitudes, and perceived survey consequentiality. Regression tables are reported in the Appendix, Table A.11

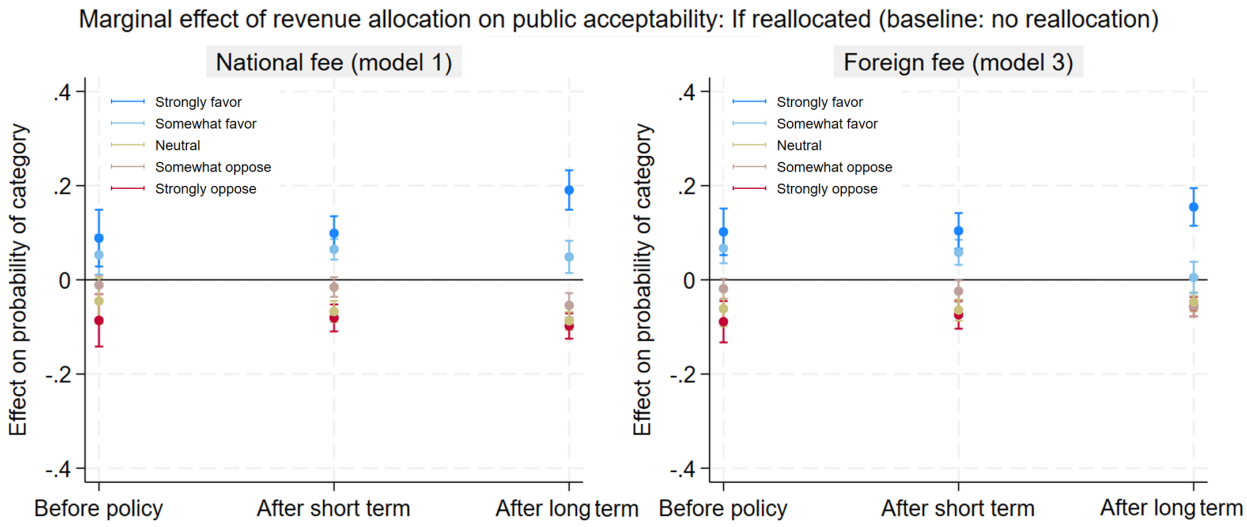

Figure 11: Marginal effects of revenues reallocation on public acceptability - Follow-up study - Respondents: Tourists

Notes: This figure reports the marginal effect of the revenue reallocation on tourists' acceptability of increasing entrance fees to the Galápagos National Park, compared to the baseline scenario (no reallocation) across three periods: before, shortly after (one-month) and long after (nine-months) policy implementation. Estimates are based on Models 1 and 3, corresponding to increases in fees for domestic and foreign tourists, respectively. Regression tables are reported in the Appendix, Table A.11.

6 Discussion and conclusion

6.1 General discussion

This study examined how aligning revenue allocation with stakeholder preferences can improve support for a highly contested policy, an increase in conservation entrance fees, and how these attitudes evolve following policy implementation. We show that, despite strong initial opposition among residents, directing revenues toward publicly preferred uses substantially increases policy acceptability. The probability of strong support rises by around 10% before the policy and reaches up to 50% nine months after implementation. These findings offer a practical pathway for improving the political feasibility of contested policies for managing and financing protected areas. Because delayed or ineffective implementation can accelerate environmental degradation, strengthening policy support is critical for reducing implementation frictions and enabling timely adjustments of nature-based tourism policies, ultimately aimed at conserving ecosystems and sustaining local livelihoods.

In the Galápagos Islands, persistent opposition among residents reflects concerns about economic impacts, institutional distrust, and perceived inequalities in tourism governance. Follow-up survey responses indicate that many residents perceived a decline in visitation consistent with their pre-policy expectations. However, observed visitation trends were heterogeneous. During the first semester of 2025, total arrivals increased by 2%, driven by a 15% rise in foreign visitors and a 15% decline in domestic visitors (Observatorio de Turismo de Galápagos, 2025). Because domestic tourists are predominantly land-based and more visible within local economies, this shift in demand composition likely shaped perceptions of overall decline despite stable aggregate arrivals.

The stability of foreign demand is consistent with survey findings showing that foreign tourists were generally more supportive of higher fees and less sensitive to revenue allocation than domestic visitors. This pattern also aligns with residents narratives describing foreign demand as relatively inelastic. From a governance perspective, sustained acceptability among residents may depend less on total arrivals and more on whether institutions credibly and transparently translate increased revenues into locally valued benefits delivered with distributive fairness.

6.2 Theoretical implications

This study advances public acceptability research by identifying and empirically testing a mechanism through which revenue allocation increases support for a policy change. Consistent with climate policy literature (Bergquist et al., 2020; Dolšak et al., 2020; Douenne & Fabre, 2020), our findings show that earmarking revenues can enhance support when benefits are tangible. This helps reconcile mixed evidence by suggesting that revenues allocation increases acceptability primarily when it addresses local priorities rather than abstract or distant outcomes (Uyduranoglu & Ozturk, 2020). In nature-based tourism settings, where residents bear indirect policy costs and perceive long-standing inequalities, institutional trust and effective redistribution of revenues also play a key role in shaping support for entrance fee policies.

A second contribution concerns the temporal dynamics of acceptability. Whereas much of the literature treats acceptability as a static attitude or focuses on implementation in the short-term (Carattini et al., 2018; Schuitema et al., 2010), our study shows that opposition can persist over long periods. In this case, persistent opposition reflects realized costs aligning with prior expectations. Nevertheless, revenue reallocation becomes increasingly effective after policy implementation, particularly among residents, where it operates as a compensatory mechanism that offsets perceived losses and enhances public benefits. Our analysis also evidences stakeholder heterogeneity. Foreign tourists remain largely unaffected by the fee increase, domestic tourists exhibit intermediate sensitivity, and residents show the strongest conditional responses to revenue allocation. Tourist responses further align with economic theory suggesting that destinations with strong market power, such as the Galápagos Islands given its high biodiversity and endemism, face relatively inelastic demand (Rosselló-Nadal & Sard, 2026).

Third, the study shows the importance of transparency and procedural legitimacy. While trust in public institutions is found positively associated with policy support as in Rodriguez-Sanchez et al. (2018), our qualitative analysis indicate that low institutional credibility is linked to perceived limitations in implementation capacity, skepticism about policy motivations, and concerns about reinforcing inequalities. Together, these findings suggest that incorporating stakeholder preferences into revenue allocation enhances acceptability, but public support extends beyond cost–benefit evaluations, involving perceptions of legitimacy and transparency in resource management as prereq-

uisites. This supports literature highlighting that bottom-up governance approaches may be more effective than top-down policy imposition, especially for policies governing tourism in protected areas (Perlaviciute & Squintani, 2020).

6.3 Practical implications

Our findings align with the argument that entrance fees are a central governance instrument in protected areas, rather than merely a financing mechanism (Mach et al., 2020). For policymakers, a key implication is that the political feasibility of entrance fee adjustments depends on transparent and credible mechanisms for regulating tourism and delivering tangible benefits to local communities.

In the Galápagos Islands, a transparent and fair redistribution of public revenues could help mitigate the political instability that has undermined conservation efforts for more than two decades (Burbano & Meredith, 2020; Castrejón & Defeo, 2023; King et al., 2025). Given residents' stated preferences, authorities should prioritize investments in public services, particularly safe water and solid waste management, alongside continued efforts in species conservation. Such investments may reduce tensions between residents and protected area managers about strategies to transition toward a zero-growth tourism model, as urged by the World Heritage Committee (World Heritage Centre, 2018). The prioritization of both public services and species conservation also suggests that residents recognize the importance of protecting fragile ecosystems on which their livelihoods depend, creating opportunities to build a shared vision across stakeholder groups. As the fee increase is expected to double revenues, efficient financial mechanisms will be critical to optimize expenditures and align them with stakeholder priorities (Consejo de Gobierno del Régimen Especial de Galápagos, 2024; Laruelle, 2021).

More broadly, strengthening governance could involve participatory budgeting processes for allocating entrance fee revenues and other funding sources. Such approaches are both normatively desirable and strategically important (King et al., 2025), as they may help restore institutional trust and facilitate implementation of other contested policies, including fishing gear restrictions and marine zoning (Castrejón & Defeo, 2023; Castrejón et al., 2024). Enhanced coordination among institutions sharing fee revenues is also essential for an effective long-term governance (Burbano et

al., 2022; King et al., 2025). Finally, if a transition toward a zero-growth tourism model remains a policy objective, complementary tools beyond fee increases, such as differentiated pricing schemes, visitor quotas, or flight restrictions, should be considered (Burbano et al., 2022). Evaluating the acceptability of these measures represents an important direction for future policy design.

6.4 Limitations and future research

Although within-subject designs are well suited for examining shifts in acceptability across policy scenarios, between-subject designs may be more appropriate for assessing entirely new policies where respondents lack familiarity with the baseline. Future research could integrate both designs to evaluate alternative revenue allocation strategies. While this study captured attitudes nine months after implementation, longer observation periods would enable assessment of how realized environmental and economic outcomes reshape acceptability beyond the first year.

Besides revenue allocation, beliefs about demand impacts were an important factor in residents' narratives. Future studies could experimentally test how information about tourism demand elasticity or observed visitation trends influences acceptability and whether beliefs mediate support, although prior evidence suggests limited mediation effects (García-Antúnez et al., 2023). Incorporating willingness-to-pay and willingness-to-accept approaches could further link acceptability to welfare measures. Finally, field experiments examining participatory budgeting or co-designed revenue allocation processes could identify institutional arrangements that are more effective for building and sustaining public support for conservation financing policies.

Ultimately, entrance fees represent only one component of broader challenges in the governance of protected areas and nature-based tourism (Mach et al., 2020). However, aligning revenue allocation with both conservation priorities and local needs offers a practical pathway for restoring legitimacy and building public support. By providing a framework for improving the acceptability of policy instruments in high-biodiversity destinations, this study contributes to debates on how to design politically feasible policies that support the governance and long-term success of protected areas.

References

Bergquist, P., Mildenberger, M., & Stokes, L. C. (2020). Combining climate, economic, and social policy builds public support for climate action in the US. *Environmental Research Letters*, *15*(5), 054019. https://doi.org/10.1088/1748-9326/ab81c1

Burbano, D. V., & Meredith, T. C. (2020). Conservation strategies through the lens of small-scale fishers in the Galapagos Islands, Ecuador: Perceptions underlying local resistance to marine planning. *Society & Natural Resources*, *33*(10), 1194–1212. https://doi.org/10.1080/08941920.2020.1765058

Burbano, D. V., Valdivieso, J. C., Izurieta, J. C., Meredith, T. C., & Ferri, D. Q. (2022). "rethink and reset" tourism in the Galapagos Islands: Stakeholders' views on the sustainability of tourism development. *Annals of Tourism Research Empirical Insights*, *3*(2), 100057. https://doi.org/10.1016/j.annale.2022.100057

Carattini, S., Baranzini, A., & Lalive, R. (2018). Is Taxing Waste a Waste of Time? Evidence from a Supreme Court Decision. *Ecological Economics*, *148*, 131–151. https://doi.org/10.1016/j.ecolecon.2018.02.001

Carson, R., & Groves, T. (2007). Incentive and informational properties of preference questions. *Environmental Resource Economics*, *37*(1), 181–210. https://EconPapers.repec.org/RePEc:kap:enreec:v:37:y:2007:i:1:p:181-210

Castrejón, M., & Defeo, O. (2023). Reconsidering the longline ban in the Galapagos Marine Reserve. *Marine Policy*, *151*, 105589. https://doi.org/10.1016/j.marpol.2023.105589

Castrejón, M., Moity, N., & Charles, A. (2024). The bumpy road to conservation: Challenges and opportunities in updating the Galapagos zoning system. *Marine Policy*, *163*, 106146. https://doi.org/10.1016/j.marpol.2024.106146

CGREG. (2015). Ley Orgánica de Régimen Especial de la Provincia de Galápagos (Registro Oficial, Segunda Edición, Suplemento 2, 11 de junio de 2015) [Accessed: 2025-09-02]. https://www.gobiernogalapagos.gob.ec/biblioteca/#

Chung, J. Y., Kyle, G. T., Petrick, J. F., & Absher, J. D. (2011). Fairness of prices, user fee policy and willingness to pay among visitors to a national forest. *Tourism Management*, *32*(5), 1038–1046. https://doi.org/10.1016/j.tourman.2010.08.016

Consejo de Gobierno del Régimen Especial de Galápagos. (2024). *Informe técnico respecto a la distribución y actualización de la tasa de ingreso por conservación de áreas naturales protegidas*. Gobierno del Régimen Especial de Galápagos. Retrieved November 5, 2025, from https://www.gobiernogalapagos.gob.ec/wp-content/uploads/downloads/2024/03/c.2.-informe_tecnico_st_-_aprobado_por_ctpt.pdf

Convention on Biological Diversity. (2022). *Target 3: Conserve 30% of land, waters and seas*. Retrieved December 2, 2025, from https://www.cbd.int/gbf/targets/3

Daly, C., Fraser, G., & Snowball, J. (2015). Willingness to pay for marine-based tourism in the Ponta do Ouro Partial Marine Reserve, Mozambique. *African Journal of Marine Science*, *37*(1), 33–40. https://doi.org/10.2989/1814232X.2015.1012556

Davidian, M., & Giltinan, D. M. (2003). Nonlinear models for repeated measurement data: An overview and update. *Journal of Agricultural, Biological, and Environmental Statistics*, *8*(4). https://doi.org/10.1198/1085711032697

Dolšak, N., Adolph, C., & Prakash, A. (2020). Policy design and public support for carbon tax: Evidence from a 2018 US national online survey experiment. *Public Administration*, *98*(4), 905–921. https://doi.org/10.1111/padm.12657

Douenne, T., & Fabre, A. (2020). French attitudes on climate change, carbon taxation and other climate policies. *Ecological Economics*, *169*, 106496. https://doi.org/10.1016/j.ecolecon.2019.106496

Dunlap, R. E., Van Liere, K. D., Mertig, A. G., & Jones, R. E. (2000). New Trends in Measuring Environmental Attitudes: Measuring Endorsement of the New Ecological Paradigm: A Revised NEP Scale. *Journal of Social Issues*, *56*(3), 425–442. https://doi.org/https://doi.org/10.1111/0022-4537.00176

El Diario. (2025, February). Corte Constitucional admite trámite de acción por inconstitucionalidad contra aumento de tarifas en Galápagos. https://www.eldiario.ec/ecuador/ecuador-corte-constitucional-admite-tramite-de-accion-por-inconstitucionalidad-contra-aumento-de-tarifas-en-galapagos-27022025/

Fuster, A., & Zafar, B. (2023). Chapter 4 - survey experiments on economic expectations. In R. Bachmann, G. Topa, & W. van der Klaauw (Eds.), *Handbook of economic expectations* (pp. 107–130). Academic Press. https://doi.org/10.1016/B978-0-12-822927-9.00010-0

García-Antúnez, O., Lampinen, J., Raymond, C. M., Gulsrud, N. M., & Olafsson, A. S. (2023). Unpacking public perceptions of carbon sequestration and storage in urban greenery: Implications for the social acceptability of carbon-oriented nature-based solutions. *Nature-Based Solutions*, *4*, 100087. https://doi.org/https://doi.org/10.1016/j.nbsj.2023.100087

Gobierno del Regimen Especial de Galapagos. (2019). Documento 2. Resolución del Pleno del Consejo de Gobierno de Galápagos sobre nueva tasa de ingreso al PNG (2108) [Accessed: 2025-10-28]. https://www.gobiernogalapagos.gob.ec/estudios-sobre-tasa-de-ingreso/

Grant, P. R., & Grant, B. R. (2008). *How and Why Species Multiply: The Radiation of Darwin's Finches*. Princeton University Press. https://doi.org/10.2307/j.ctvx5wc23

INEC. (2023). *Censo Ecuador - Resultados*. Retrieved November 13, 2024, from https://www.censoecuador.gob.ec/resultados-censo/

INEC. (2024). *Sistema de Estadísticas Laborales y Empresariales*. Retrieved November 13, 2024, from https://www.ecuadorencifras.gob.ec/sistema-estadisticas-laborales-empresariales/

Kallbekken, S., & Sælen, H. (2011). Public acceptance for environmental taxes: Self-interest, environmental and distributional concerns. *Energy Policy*, *39*(5), 2966–2973. https://doi.org/https://doi.org/10.1016/j.enpol.2011.03.006

King, C., Hunt, C., Barragán-Paladines, M. J., Muñoz-Barriga, A., Maria, V. S., Díaz, S. C., Norris, L., Reyes, N. W., Bensted-Smith, R., & Sandbrook, C. (2025). Navigating the Galápagos paradox: Tourism growth management discourses in protected areas. *Tourism Geographies*, 1–26. https://doi.org/10.1080/14616688.2025.2582669

Laruelle, A. (2021). Voting to select projects in participatory budgeting. *European Journal of Operational Research*, *288*(2), 598–604. https://doi.org/https://doi.org/10.1016/j.ejor.2020.05.063

Mach, L., Winner, C., Rojas, C., & Klemond, M. (2020). Protected area entry fees and governance quality. *Tourism Management*, *77*, 104003. https://doi.org/10.1016/j.tourman.2019.104003

Mmopelwa, G., Kgathi, D., & Molefhe, L. (2007). Tourists' perceptions and their willingness to pay for park fees: A case study of self-drive tourists and clients for mobile tour operators in Moremi Game Reserve, Botswana. *Tourism Management*, *28*(4), 1044–1056. https://doi.org/10.1016/j.tourman.2006.08.014

Monz, C. A., Pickering, C. M., & Hadwen, W. L. (2013). Recent advances in recreation ecology and the implications of different relationships between recreation use and ecological impacts. *Frontiers in Ecology and the Environment*, *11*(8), 441–446. https://doi.org/https://doi.org/10.1890/120358

Nature. (1935). Darwin and the Galápagos Islands. *Nature*, *136*, 506. https://doi.org/10.1038/136506b0

Observatorio de Turismo de Galápagos. (2025). *Arribos mensuales*. Retrieved August 19, 2025, from https://www.observatoriogalapagos.gob.ec/

Perlaviciute, G., & Squintani, L. (2020). Public participation in climate policy making: Toward reconciling public preferences and legal frameworks. *One Earth*, *2*(4), 341–348. https://doi.org/https://doi.org/10.1016/j.oneear.2020.03.009

Pizzitutti, F., Walsh, S. J., Rindfuss, R. R., Gunter, R., Quiroga, D., Tippett, R., & Mena, C. F. (2017). Scenario planning for tourism management: a participatory and system

dynamics model applied to the Galapagos Islands of Ecuador. *Journal of Sustainable Tourism*, *25*(8), 1117–1137. https://doi.org/10.1080/09669582.2016.1257011

Rodriguez-Sanchez, C., Schuitema, G., Claudy, M., & Sancho-Esper, F. (2018). How trust and emotions influence policy acceptance: The case of the Irish water charges. *British Journal of Social Psychology*, *57*(3), 610–629. https://doi.org/10.1111/bjso.12242

Rosselló-Nadal, J., & Sard, M. (2026). Tourism taxation: Balancing revenues, competitiveness and sustainability in destination management. *Tourism Management*, *113*, 105326. https://doi.org/https://doi.org/10.1016/j.tourman.2025.105326

Schuitema, G., Steg, L., & Forward, S. (2010). Explaining differences in acceptability before and acceptance after the implementation of a congestion charge in Stockholm. *Transportation Research Part A: Policy and Practice*, *44*(2), 99–109. https://doi.org/10.1016/j.tra.2009.11.005

The Guardian. (2024, August). The problem with people: How more tourists and a growing population are taking their toll on the galápagos islands. https://www.theguardian.com/global-development/article/2024/aug/15/dying-species-and-garbage-the-downsides-of-the-galapagos-tourism-dependency?CMP=share_btn_url

The New York Times. (2019, September). Sharp Increase in Visitor Park Fees Proposed for the Galápagos. https://www.nytimes.com/2019/09/26/travel/galpagos-island-park-fees.html

Uyduranoglu, A., & Ozturk, S. S. (2020). Public support for carbon taxation in Turkey: Drivers and barriers. *Climate Policy*, *20*(9), 1175–1191. https://doi.org/10.1080/14693062.2020.1816887

Williams, R., & Quiroz, C. (2020). *Ordinal regression models*. SAGE Publications Ltd. https://doi.org/10.4135/9781526421036885901

World Heritage Centre. (2007). State of Conservation: Galápagos Islands (Ecuador), 2007 [Accessed: 2025-09-02]. https://whc.unesco.org/en/soc/994

World Heritage Centre. (2018). Decision 42 COM 7B.85 (Galápagos Islands) [Accessed: 2025-08-20]. https://whc.unesco.org/en/decisions/7314/

Wüstenhagen, R., Wolsink, M., & Bürer, M. J. (2007). Social acceptance of renewable energy innovation: An introduction to the concept. *Energy Policy*, *35*(5), 2683–2691. https://doi.org/https://doi.org/10.1016/j.enpol.2006.12.001

Zhang, C., Ma, B., & Du, Q. (2023). Public acceptability and its determinants of unit pricing for municipal solid waste disposal: Evidence from a household survey in Beijing. *Journal of Environmental Management*, *346*, 118966. https://doi.org/10.1016/j.jenvman.2023.118966

Zhu, X., & Lu, C. (2017). Re-evaluation of the New Ecological Paradigm scale using item response theory. *Journal of Environmental Psychology*, *54*, 79–90. https://doi.org/10.1016/j.jenvp.2017.10.005

Zou, S. S., Pitas, N. A., Cho, S. J., & Yoon, H. V. (2024). The moderating effect of place attachment on visitors' trust and support for recreational fees in national parks. *Journal of Environmental Psychology*, *98*, 102412. https://doi.org/10.1016/j.jenvp.2024.102412

A Appendix

Table A.1: Sample characteristics - Local Residents - Balanced sample

Variable		Before policy change	After policy one-month	After policy nine-month
Residence Status	Non-permanent	16%	15%	10%
	Permanent	84%	85%	90%
Gender	Male	53%	53%	53%
	Female	47%	47%	47%
Age 18 - 70 years	Mean	34.7	34.7	34.7
	(std. dev.)	(12.6)	(10.6)	(10.8)
Education level	Secondary or lower	64%	64%	64%
	Higher education	36%	36%	36%
Monthly income USD 0.25k - 6k	Mean	0.89k	0.89k	0.89k
	(std. dev.)	(0.97)	(0.65)	(0.50)
Economic sector	Commerce	16%	15%	19%
	Tourism-related	47%	48%	43%
	Govt. or conservation	12%	12%	12%
	Other sectors	14%	14%	14%
	Students and none	12%	12%	12%
Policy change awareness	Yes	96%	99%	89%
	No	4%	1%	11%
Observations		215	215	215

Notes: The samples after policy change are weighted to match the characteristics of the sample before policy (1 month); weights are calculated using entropy balance based on age, gender, income, education and economic sector.

Table A.2: Sample characteristics - Tourists - Balanced sample

Variable		Before policy change	After policy one-month	After policy nine-months
Tourist nationality	Domestic	42%	42%	42%
	Foreign	58%	58%	58%
Gender	Male	50%	50%	50%
	Female	40%	50%	50%
Age	Mean	41.1	41.1	41.1
18 - 83 years	(std. dev.)	(15.9)	(15.0)	(13.7)
Education	Secondary or lower	12%	12%	12%
	Higher education	88%	88%	88%
Monthly income	Mean	10.5k	10.5k	10.5k
USD 0.3k - 150k	(std. dev.)	(26.1k)	(25.5k)	(27.4k)
Occupation	Private worker	53%	57%	62%
	Public employee	26%	22%	24%
	Retired	12%	8%	7%
	Other	9%	12%	7%
Tourist modality	Land-based	67%	67%	67%
	Liveaboard	33%	33%	33%
Lenght of stay	Mean	8.9	7.0	7.2
3 - 60 days	(std. dev.)	(10.8)	(4.6)	(5.9)
Policy change awareness	Yes	46%	60%	38%
	No	54%	40%	62%
Observations		376	376	376

Notes: The samples before policy change and after policy (1 month) are weighted to match the characteristics of the sample after policy (9 months); weights are calculated using entropy balance based on nationality, age, education, income and visit modality.

Table A.3: Public acceptability - Pearson's chi-squared test: Without revenues reallocation vs If revenues are reallocated - Respondents: Residents

Public acceptability 5-point Likert scale	National fee increase		Difference %	Pearson's chi-squared Ho: No difference
	W/o funds reallocation	If funds were reallocated		
Strongly oppose	51%	16%	-35%	chi2 = 326.60
Somewhat oppose	16%	8%	-8%	p = 0.000
Neutral	17%	10%	-7%	
Somewhat favor	9%	27%	18%	
Strongly favor	7%	39%	32%	
Public acceptability 5-point Likert scale	**Foreign fee increase**		**Difference %**	**Pearson's chi-squared Ho: No difference**
	W/o funds reallocation	If funds were reallocated		
Strongly oppose	49%	16%	-33%	chi2 = 261.94
Somewhat oppose	15%	8%	-7%	p = 0.000
Neutral	14%	10%	-4%	
Somewhat favor	12%	27%	-5%	
Strongly favor	10%	39%	29%	
Observations	629	629	-	-

Table A.4: Public acceptability - Pearson's chi-squared test: Before policy change vs After short-term vs After long-term - Respondents: Residents

Public acceptability 5-point Likert scale	National fee increase w/o funds reallocation		Difference %	Pearson's chi-squared Ho: No difference
	Before policy	After short-term		
Strongly oppose	48%	48%	0%	chi2 = 4.992
Somewhat oppose	16%	24%	8%	p = 0.288
Neutral	17%	13%	-4%	
Somewhat favor	11%	9%	-3%	
Strongly favor	8%	7%	-1%	
Public acceptability 5-point Likert scale	**Foreign fee increase w/o funds reallocation**		**Difference %**	**Pearson's chi-squared Ho: No difference**
	Before policy	After short-term		
Strongly oppose	49%	40%	-9%	chi2 = 4.961
Somewhat oppose	17%	20%	3%	p = 0.291
Neutral	11%	14%	3%	
Somewhat favor	12%	16%	4%	
Strongly favor	11%	10%	-1%	
Public acceptability 5-point Likert scale	**Overall fee increase if funds were reallocated**		**Difference %**	**Pearson's chi-squared Ho: No difference**
	Before policy	After short-term		
Strongly oppose	29%	16%	-13%	chi2 = 27.215
Somewhat oppose	11%	7%	-4%	p = 0.000
Neutral	12%	14%	2%	
Somewhat favor	31%	25%	-6%	
Strongly favor	17%	38%	21%	
Observations	203	210	-	-
Public acceptability 5-point Likert scale	**National fee increase w/o funds reallocation**		**Difference %**	**Pearson's chi-squared Ho: No difference**
	After short-term	After long-term		
Strongly oppose	48%	56%	8%	chi2 = 22.281
Somewhat oppose	24%	8%	-16%	p = 0.000
Neutral	13%	21%	8%	
Somewhat favor	8%	7%	-1%	
Strongly favor	7%	8%	1%	
Public acceptability 5-point Likert scale	**Foreign fee increase w/o funds reallocation**		**Difference %**	**Pearson's chi-squared Ho: No difference**
	After short-term	After long-term		
Strongly oppose	40%	58%	18%	chi2 = 25.132
Somewhat oppose	20%	7%	-13%	p = 0.000
Neutral	14%	18%	4%	
Somewhat favor	16%	8%	-8%	
Strongly favor	10%	9%	-1%	
Public acceptability 5-point Likert scale	**Overall fee increase if funds were reallocated**		**Difference %**	**Pearson's chi-squared Ho: No difference**
	After short-term	After long-term		
Strongly oppose	16%	5%	-11%	chi2 = 37.844
Somewhat oppose	7%	6%	-1%	p = 0.000
Neutral	14%	4%	-10%	
Somewhat favor	25%	24%	-1%	
Strongly favor	38%	61%	23%	
Observations	210	216	-	-

Table A.5: Public acceptability - Pearson's chi-squared test: W/o funds reallocation vs If funds are reallocated - Respondents: Tourists

National Tourists				
Public acceptability 5-point Likert scale	National fee increase		Difference %	Pearson's chi-squared Ho: No difference
	W/o funds reallocation	If funds were reallocated		
Strongly oppose	24%	11%	-13%	chi2 = 72.296
Somewhat oppose	24%	10%	-14%	p = 0.000
Neutral	23%	23%	0%	
Somewhat favor	17%	26%	9%	
Strongly favor	12%	30%	18%	
Observations	365	364	-	-
Foreign Tourists				
Public acceptability 5-point Likert scale	Foreign fee increase		Difference %	Pearson's chi-squared Ho: No difference
	W/o funds reallocation	If funds were reallocated		
Strongly oppose	10%	4%	-6%	chi2 = 26.763
Somewhat oppose	14%	7%	-7%	p = 0.000
Neutral	20%	19%	-1%	
Somewhat favor	25%	25%	0%	
Strongly favor	31%	45%	14%	
Observations	363	361	-	-

Table A.6: Public acceptability - Pearson's chi-squared test: Before policy vs After short-term vs After long-term - Respondents: National tourists

Domestic tourists				
Public acceptability	National fee increase w/o funds reallocation		Difference	Pearson's chi-squared
5-point Likert scale	Before policy	After short-term	%	Ho: No difference
Strongly oppose	43%	22%	-21%	chi2 = 11.220
Somewhat oppose	19%	19%	0%	p = 0.024
Neutral	12%	25%	13%	
Somewhat favor	14%	19%	5%	
Strongly favor	12%	15%	3%	
Public acceptability	Overall fee increase if funds were reallocated		Difference	Pearson's chi-squared
5-point Likert scale	Before policy	After short-term	%	Ho: No difference
Strongly oppose	20%	9%	-11%	chi2 = 10.468
Somewhat oppose	18%	10%	-8%	p = 0.033
Neutral	16%	29%	13%	
Somewhat favor	24%	23%	-1%	
Strongly favor	22%	29%	7%	
Public acceptability	National fee increase w/o funds reallocation		Difference	Pearson's chi-squared
5-point Likert scale	After short-term	After long-term	%	Ho: No difference
Strongly oppose	22%	17%	-5%	chi2 = 6.322
Somewhat oppose	19%	29%	10%	p = 0.176
Neutral	24%	28%	4%	
Somewhat favor	19%	16%	-3%	
Strongly favor	15%	10%	-5%	
Public acceptability	Overall fee increase if funds were reallocated		Difference	Pearson's chi-squared
5-point Likert scale	After short-term	After long-term	%	Ho: No difference
Strongly oppose	9%	8%	-1%	chi2 = 3.562
Somewhat oppose	10%	7%	-3%	p = 0.469
Neutral	29%	22%	-7%	
Somewhat favor	23%	28%	5%	
Strongly favor	29%	35%	6%	
Observations	205	290	-	-

Table A.7: Public acceptability - Pearson's chi-squared test: Before policy vs After short-term vs After long-term - Respondents: Foreign tourists

Foreign tourists				
Public acceptability	Foreign fee increase w/o funds reallocation		Difference	Pearson's chi-squared
5-point Likert scale	Before policy	After short-term	%	Ho: No difference
Strongly oppose	4%	17%	13%	chi2 = 6.485
Somewhat oppose	20%	15%	-5%	p = 0.166
Neutral	26%	19%	-7%	
Somewhat favor	20%	26%	6%	
Strongly favor	30%	23%	-7%	
Public acceptability	Overall fee increase if funds were reallocated		Difference	Pearson's chi-squared
5-point Likert scale	Before policy	After short-term	%	Ho: No difference
Strongly oppose	4%	5%	1%	chi2 = 3.431
Somewhat oppose	5%	10%	5%	p = 0.488
Neutral	25%	21%	-4%	
Somewhat favor	23%	32%	9%	
Strongly favor	43%	32%	-11%	
Public acceptability	Foreign fee increase w/o funds reallocation		Difference	Pearson's chi-squared
5-point Likert scale	After short-term	After long-term	%	Ho: No difference
Strongly oppose	17%	7%	-10%	chi2 = 10.991
Somewhat oppose	15%	12%	-3%	p = 0.027
Neutral	19%	19%	0%	
Somewhat favor	26%	26%	0%	
Strongly favor	23%	36%	13%	
Public acceptability	Overall fee increase if funds were reallocated		Difference	Pearson's chi-squared
5-point Likert scale	After short-term	After long-term	%	Ho: No difference
Strongly oppose	5%	3%	-2%	chi2 = 11.521
Somewhat oppose	10%	6%	-4%	p = 0.021
Neutral	21%	17%	-4%	
Somewhat favor	32%	22%	-10%	
Strongly favor	32%	52%	10%	
Observations	143	317	-	-

Table A.8: Mixed Effects Ordered Logit - Public Acceptability of increasing conservation entry fees - Respondents: Residents

Dependent variable: Public acceptability 5-point Likert scale	(1) National fee increase	(2) National fee increase	(3) Foreign fee increase	(4) Foreign fee increase
Policy change × Reallocation				
Before policy, no reallocation	–	–	–	–
After short-term	-0.170 (0.329)	-0.272 (0.309)	0.555 (0.314)*	0.491 (0.305)
If reallocated	1.770 (0.218)***	1.746 (0.219)***	1.583 (0.217)***	1.580 (0.221)***
After short-term × If reallocated	1.707 (0.325)***	1.724 (0.331)***	0.854 (0.301)***	0.846 (0.307)***
Gender				
Male		–		–
Female		-0.190 (0.295)		-0.222 (0.279)
Education				
Secondary or lower		–		–
Higher education		-0.188 (0.294)		-0.093 (0.273)
Economic sector				
Other sectors		–		–
Tourism and commerce		-0.245 (0.342)		-0.289 (0.327)
Govt. and conservation		1.901 (0.515)***		1.644 (0.508)***
Students and none		0.880 (0.524)*		0.827 (0.475)*
Age		-0.242 (0.150)		-0.151 (0.140)
Income		0.678 (0.234)***		0.581 (0.229)**
Trust in public inst.		0.492 (0.155)***		0.484 (0.145)***
Env. Attitude		0.380 (0.231)*		0.426 (0.208)**
Consequentiality		0.110 (0.282)		-0.054 (0.266)
Var(μ_i)	6.262 (1.178)***	4.966 (0.982)***	5.041 (0.973)***	4.206 (0.832)***
Observations	822.	810	822	810
Log-likelidhood	-1136.821	-1089.518	-1168.457	-1122.144
AIC	2289.641	2215.036	2352.914	2280.287
BIC	2327.335	2299.583	2390.608	2364.834

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses. The sample before policy change has been weighted to match the characteristics of the sample after one month; weights are calculated using entropy balance based on age, gender, income, education and economic sector.

Table A.9: Mixed Effects Ordered Logit - Public Acceptability of increasing conservation entry fees - Respondents: Tourists

Dependent variable: Public acceptability 5-point Likert scale	(1) National fee increase	(2) National fee increase	(3) Foreign fee increase	(4) Foreign fee increase
Policy change × Reallocation				
Before policy, no reallocation	–	–	–	–
After short-term	0.622 (0.499)	0.166 (0.463)	0.519 (0.478)	0.147 (0.467)
If reallocated	1.469 (0.362)***	1.320 (0.369)***	1.204 (0.321)***	1.033 (0.324)**
After short-term × If reallocated	-0.011 (0.427)	0.134 (0.435)	-0.116 (0.378)	0.038 (0.382)
Nationality × Tourist modality				
Foreign × Land-based		–		–
Foreign × Liveaboard		1.187 (0.628)		1.189 (0.647)
Domestic × Land-based		-1.902 (0.473)***		-0.877 (0.476)
Domestic × Liveaboard		0.453 (0.824)		0.928 (0.765)
Gender				
Male		–		–
Female		0.220 (0.379)		0.349 (0.364)
Education				
Secondary or lower		–		–
Higher education		1.244 (0.522)*		1.444 (0.509)**
Age		0.012 (0.013)		0.015 (0.014)
Income		-0.009 (0.007)		-0.013 (0.006)*
Env. Attitude		0.453 (0.172)**		0.393 (0.151)**
Consequentiality		1.247 (0.494)*		1.147 (0.488)*
Var(μ_i)	11.267 (2.093)***	9.247 (1.771)***	9.966 (1.939)***	8.279 (1.655)***
Observations	669	652	672	654
Log-likelihood	-1141.714	-1066.767	-1165.216	-1100.696
AIC	2299.429	2167.533	2346.431	2235.392
BIC	2335.475	2243.694	2382.513	2311.605

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses. The sample before policy change has been weighted to match the characteristics of the sample after one month; weights are calculated using entropy balance based on nationality, visit modality, gender, age and income.

Table A.10: Mixed Effects Ordered Logit - Public Acceptability of increasing conservation entry fees - Respondents: Residents (long-term results)

Dependent variable: Public acceptability 5-point Likert scale	(1) National fee increase	(2) National fee increase	(3) Foreign fee increase	(4) Foreign fee increase
Policy change × Reallocation				
Before policy, no reallocation	–	–	–	–
After short-term	-0.175 (0.277)	-0.263 (0.266)	0.449 (0.272)*	0.417 (0.268)
After long-term	-0.277 (0.277)	-0.778 (0.294)***	-0.250 (0.283)	-0.742 (0.299)**
If reallocated	1.361 (0.187)***	1.349 (0.190)***	1.290 (0.203)***	1.297 (0.207)***
After short-term × If reallocated	1.546 (0.301)***	1.573 (0.307)***	0.788 (0.291)***	0.770 (0.297)***
After long-term × If reallocated	3.053 (0.357)***	3.121 (0.364)***	2.879 (0.347)***	2.965 (0.356)***
Gender				
Male		–		–
Female		-0.046 (0.182)		-0.062 (0.178)
Education				
Secondary or lower		–		–
Higher education		-0.086 (0.189)		-0.043 (0.185)
Economic sector				
Other sectors		–		–
Tourism and commerce		0.054 (0.226)		-0.076 (0.221)
Govt. and conservation		1.908 (0.372)***		1.655 (0.371)***
Students and none		0.564 (0.332)*		0.768 (0.331)**
Age		-0.138 (0.093)		-0.023 (0.092)
Income		0.433 (0.179)**		0.441 (0.179)**
Trust in public inst.		0.309 (0.100)***		0.322 (0.100)***
Env. Attitude		0.453 (0.110)***		0.513 (0.098)***
Consequentiality		-0.029 (0.189)		-0.004 (0.183)
Past respondent		1.330 (0.531)**		1.049 (0.519)**
var(μ_i)	2.925 (0.577)***	2.261 (0.466)***	2.550 (0.525)***	2.074 (0.431)***
Observations	1252	1240	1252	1240
Log-likelihood	-1709.008	-1642.594	-1739.215	-1672.732
AIC	3438.016	3327.188	3498.429	3387.465
BIC	3489.341	3434.768	3549.754	3495.045

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses. The samples before policy change and after one month have been weighted to match the characteristics of the sample after nine months; weights are calculated using entropy balance based on age, gender, income, education and economic sector.

Table A.11: Mixed Effects Ordered Logit - Public Acceptability of increasing conservation entry fees - Respondents: Tourists (long-term results)

Dependent variable: Public acceptability 5-point Likert scale	(1) National fee increase	(2) National fee increase	(3) Foreign fee increase	(4) Foreign fee increase
Policy change × Reallocation				
Before policy, no reallocation	–	–	–	–
After short-term	0.566 (0.517)	0.303 (0.478)	0.483 (0.507)	0.275 (0.500)
After long-term	0.848 (0.489)*	0.067 (0.443)	1.234 (0.482)**	0.612 (0.467)
If reallocated	1.274 (0.434)***	1.129 (0.455)**	1.420 (0.356)***	1.264 (0.372)***
After short-term × If reallocated	0.096 (0.490)	0.212 (0.507)	-0.048 (0.408)	0.081 (0.421)
After long-term × If reallocated	0.934 (0.482)*	1.086 (0.500)**	0.195 (0.387)	0.334 (0.400)
Nationality × Tourist modality				
Foreign × Land-based		–		–
Foreign × Liveaboard		1.384 (0.382)***		1.523 (0.409)***
Domestic × Land-based		-1.401 (0.308)***		-0.677 (0.313)**
Domestic × Liveaboard		0.098 (0.489)		0.561 (0.472)
Gender				
Male		–		–
Female		0.127 (0.255)		0.292 (0.259)
Education				
Secondary or lower		–		–
Higher education		1.433 (0.369)***		1.384 (0.370)***
Age		0.026 (0.009)***		0.021 (0.010)**
Income		0.006 (0.006)		0.001 (0.006)
Env. Attitude		0.733 (0.143)***		0.734 (0.139)***
Consequentiality		1.076 (0.312)***		0.811 (0.329)**
Var(μ_i)	11.766 (1.534)***	8.890 (1.179)***	11.423 (1.615)***	9.083 (1.294)***
Observations	1421.000	1404.000	1424.000	1406.000
Log-likelihood	-2710.524	-2551.322	-2713.636	-2584.147
AIC	5441.049	5140.644	5447.273	5206.293
BIC	5493.640	5240.338	5499.885	5306.015

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses. The samples before policy change and after one month have been weighted to match the characteristics of the sample after policy nine months; weights are calculated using entropy balance based on nationality, visit modality, gender, age and income.